



Ain Shams University

Dept. of Electronics and Communications Engineering

ECE 354s – Digital Signal Processing Basics -
Spring 2026

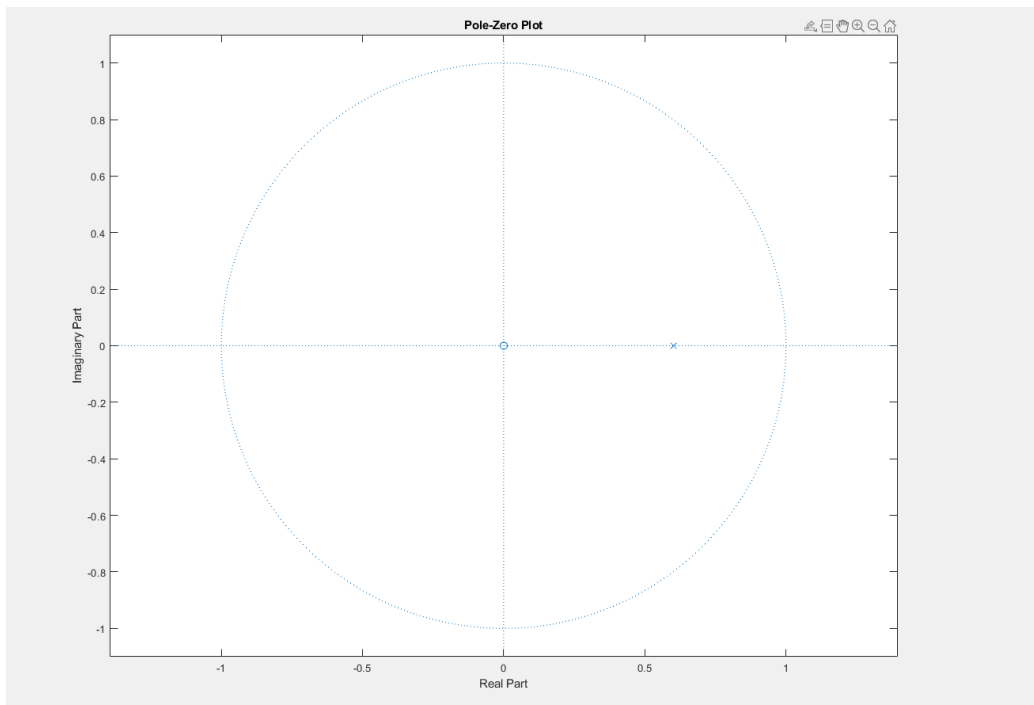
Simple Digital Filter Design Using Pole-Zero Placement Project

Made By:

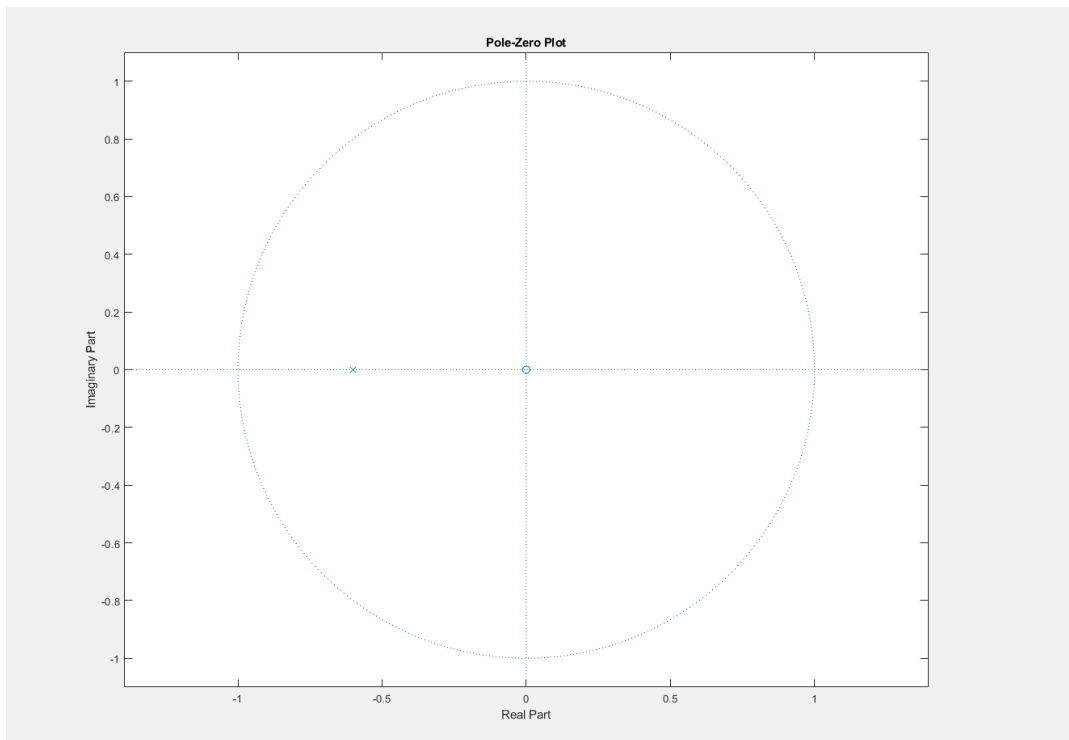
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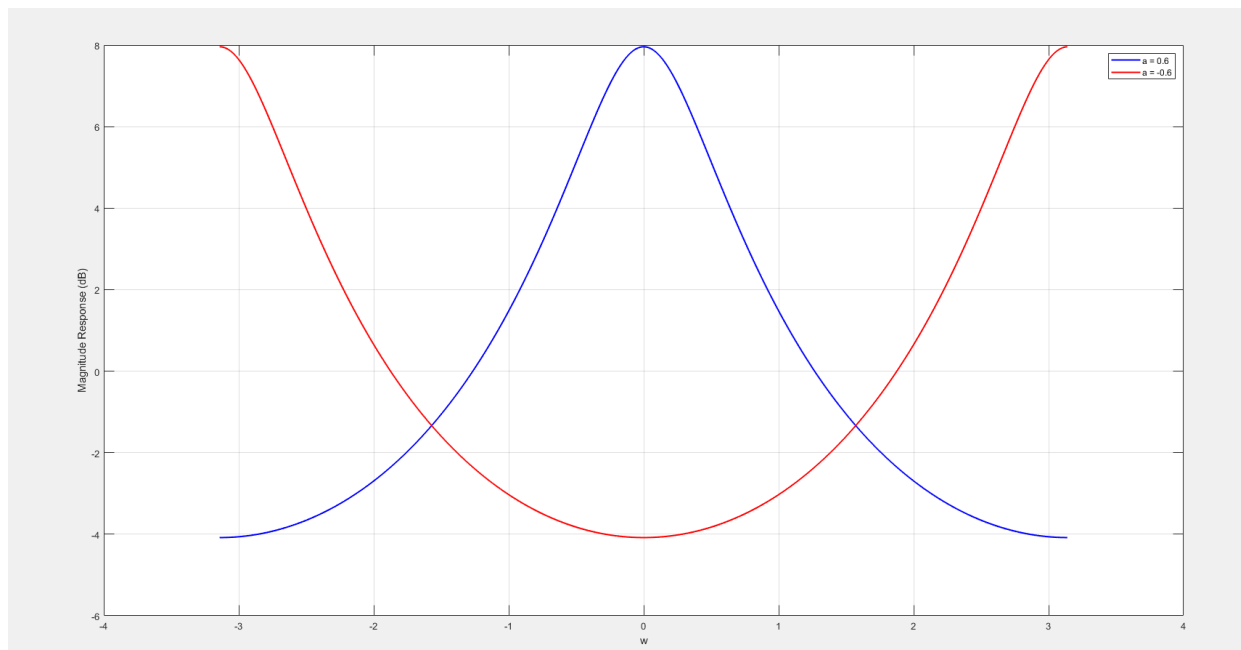
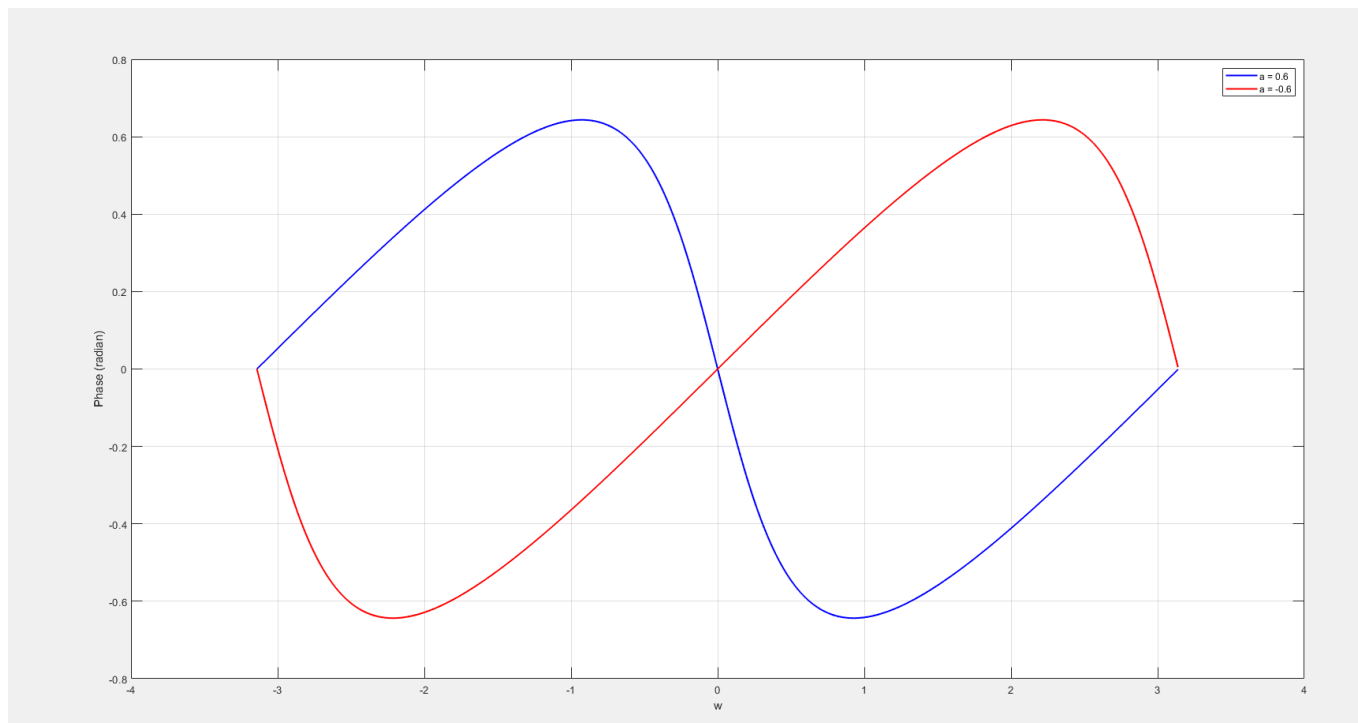
First order discrete-time low-pass filter

Pole-zero plot and the unit circle in the Z-domain for $a = 0.6$

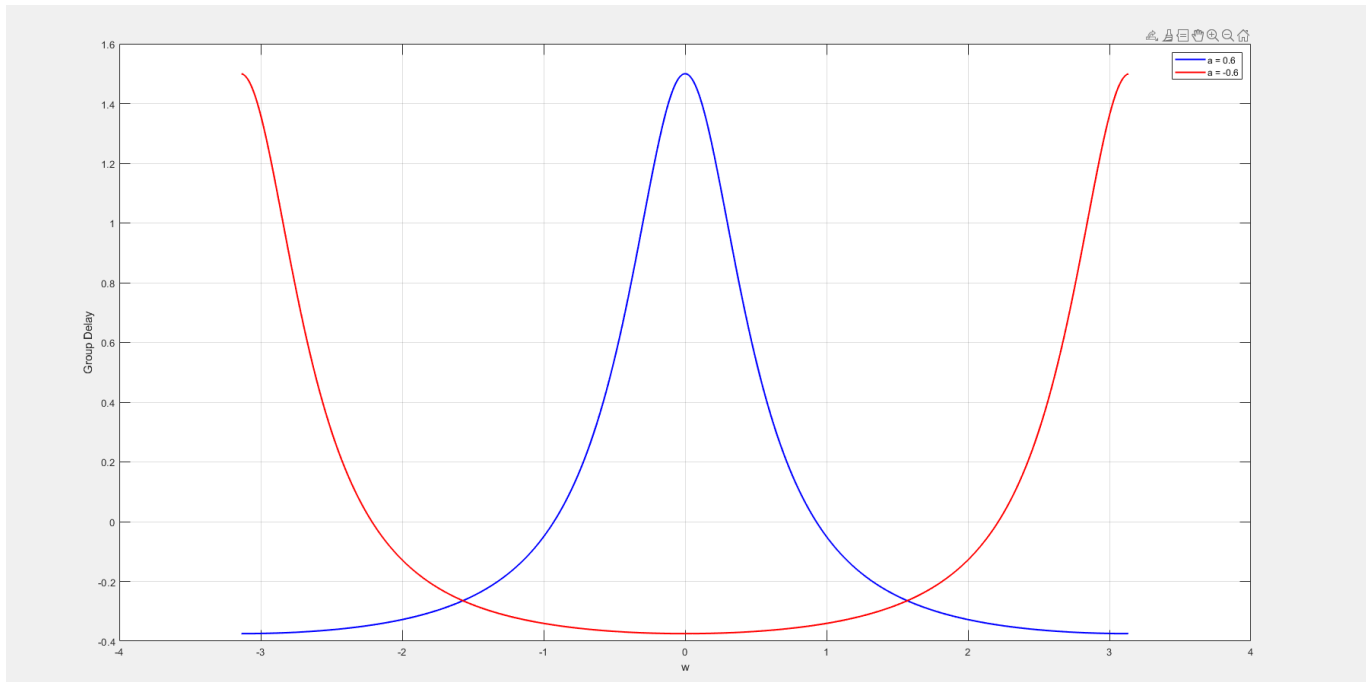


Pole-zero plot and the unit circle in the Z-domain for $a = -0.6$

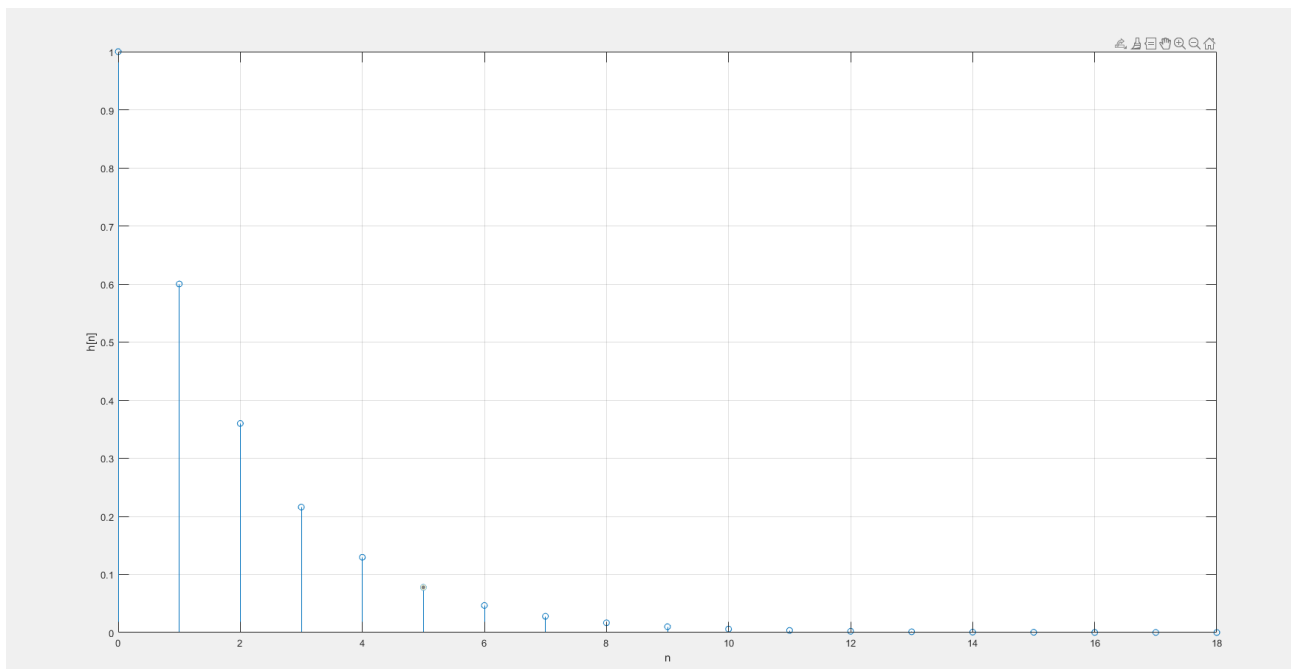


Magnitude response of the filter in dB in the frequency range $(-\pi < \omega \leq \pi)$ **Phase response of the filter in the frequency range $(-\pi < \omega \leq \pi)$** 

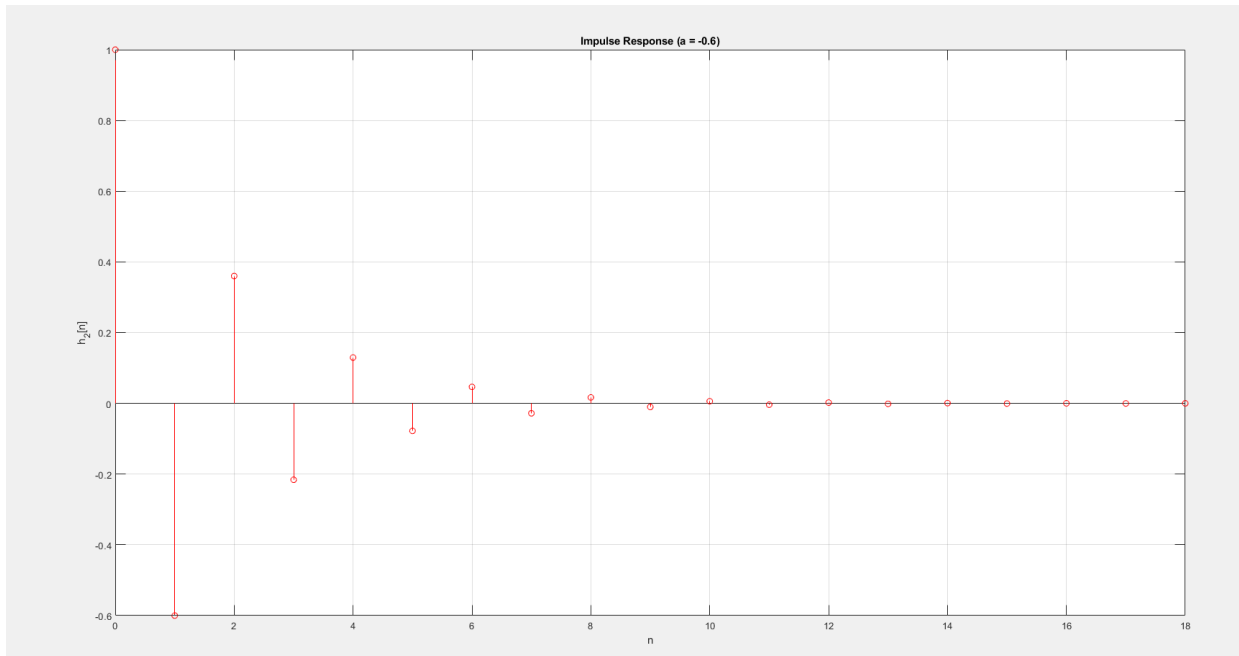
Group delay of the filter in the frequency range $(-\pi < \omega \leq \pi)$



The filter's impulse response for $a = 0.6$



The filter's impulse response for $a = -0.6$



$a = 0.6 \rightarrow$ Lowpass Filter (LPF)

The pole is located at $Z = 0.6$ on the positive real axis, which corresponds to frequency $\omega = 0$. The pole amplifies low frequencies and attenuates high frequencies $\rightarrow$ **Lowpass Filter**

$a = -0.6 \rightarrow$ Highpass Filter (HPF)

The pole is located at $Z = -0.6$ on the negative real axis, which corresponds to frequency $\omega = \pi$. The pole amplifies high frequencies and attenuates low frequencies $\rightarrow$ **High-pass Filter**

Third order discrete-time low-pass filter

The transfer function for third order filter $H(z) = \frac{(1-e^{jws}z^{-1})(1-e^{-jws}z^{-1})}{(1-p_1 \times z^{-1})(1-p_2 \times z^{-1})(1-p_3 \times z^{-1})}$

Where $p_1 = 0.6$, $p_2 = r e^{jwp}$, $p_3 = r e^{-jwp}$ and 2 complex zeros on the unit circle

A snippet from the Code showing the Sweeping on r and very small sweep on wp to get passband ripples less than 1dB

```

102 %% Sweep on r and wp
103
104 r_values = 0.1 : 0.01 : 0.99;          % Range of r
105
106 wp1_values = 0.18*pi : 0.01*pi : 0.22*pi; % Small sweep around 0.2*pi
107
108 optimum_r = 0;
109 optimum_wp1 = 0;
110
111 Optimum_ripple = inf;
112
113 for wp1 = wp1_values
114     for r_trial = r_values
115         % Complex poles
116         p2 = r_trial * exp(1j*wp1);
117         p3 = r_trial * exp(-1j*wp1);
118
119         % Complex zeros
120         z2 = exp(1j*ws0);
121         z3 = exp(-1j*ws0);
122
123         poles_vec = [p1; p2; p3];
124         zeros_vec = [z2; z3];
125
126         % Transfer function coefficients
127         b = real(poly(zeros_vec));
128         a = real(poly(poles_vec));
129
130         % Frequency response in passband
131         w_pass = linspace(0, wpo, 1000);
132
133         H_trial = freqz(b, a, w_pass);
134
135         % Magnitude in dB
136         Hdb = 20*log10(abs(H_trial));
137
138         % Passband ripple
139         ripple = max(Hdb) - min(Hdb);
140
141         % Save best parameters
142         if ripple < Optimum_ripple
143             Optimum_ripple = ripple;
144             optimum_r = r_trial;
145             optimum_wp1 = wp1;
146         end
147     end
148 end
149
150 %% optimum Parameters
151
152 fprintf('==== Optimum Parameters ==== \n');
153 fprintf('optimum r = %.3f \n', optimum_r);
154 fprintf('optimum wp1 = %.3f*pi \n', optimum_wp1/pi);
155 fprintf('Passband Ripple = %.3f dB \n', Optimum_ripple);
156

```

Command Window

```

==== Optimum Parameters ====
optimum r = 0.880
optimum wp1 = 0.210*pi
Passband Ripple = 0.530 dB

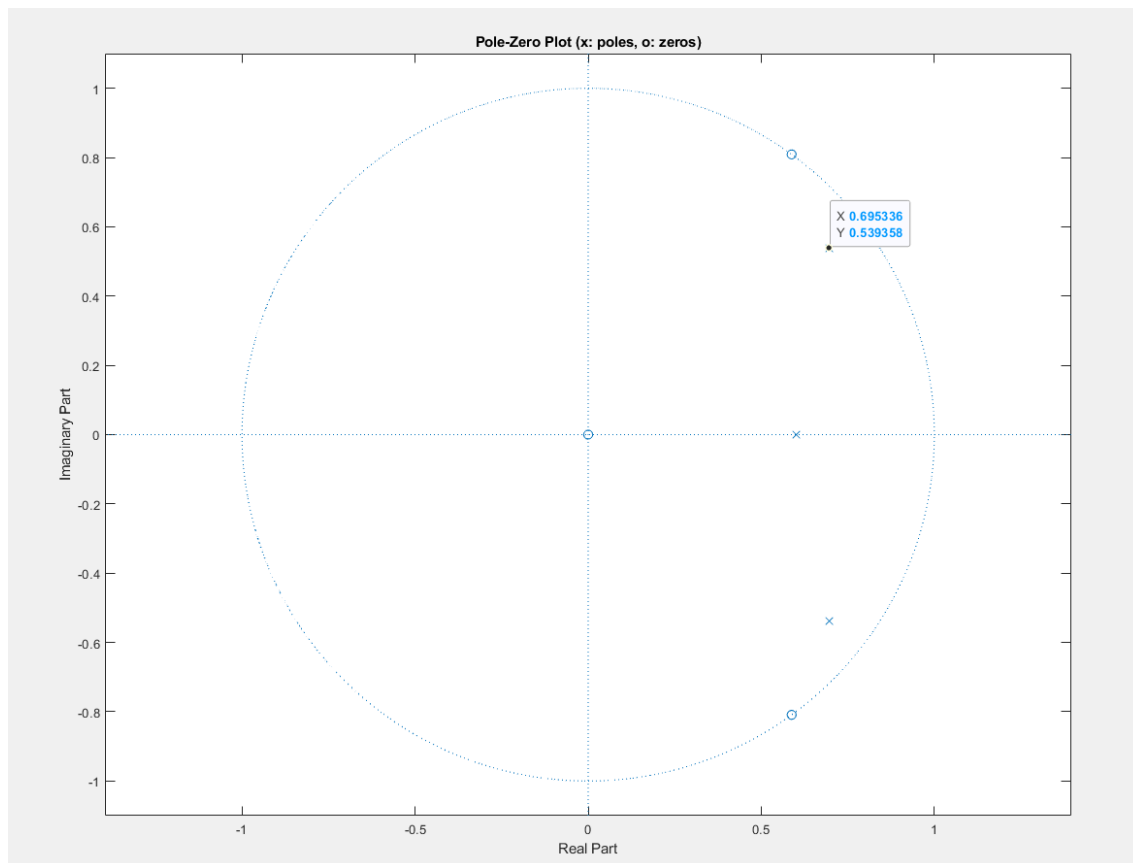
```

 >>

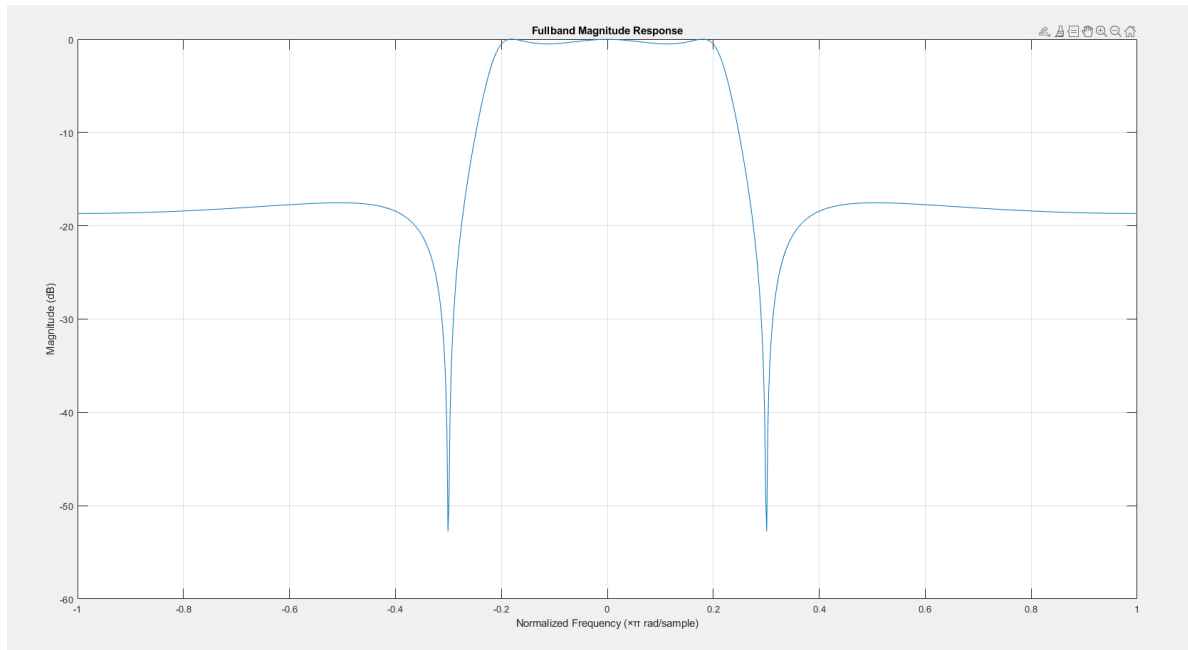
Sweeping only r will provide passband ripples of ≈ 1.1 dB (minimum), so sweeping slightly w_p around 0.2π will absolutely get passband ripples less than 1 dB, which meets the specifications of the filter.

Note: Run section of this snippet code in MATLAB file to see the results, and Full MATLAB Code is provided in the appendix

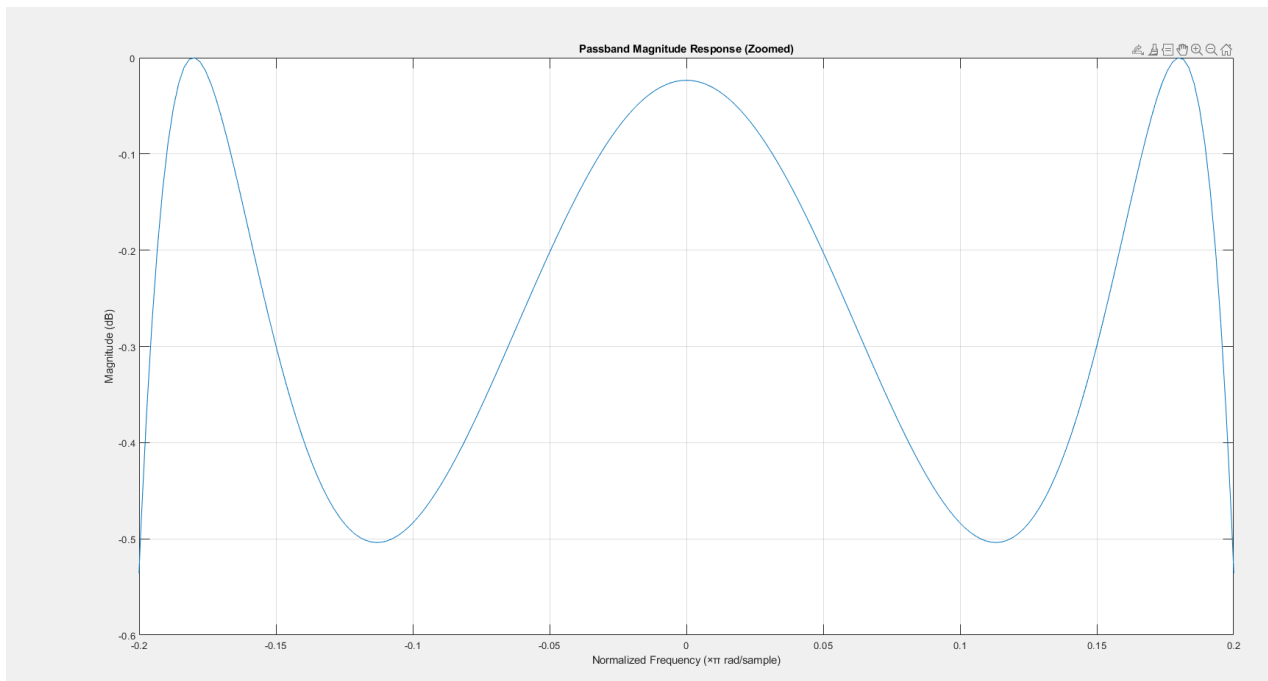
Pole-zero plot and the unit circle in the Z-domain



Magnitude response of the filter in dB in the frequency range $(-\pi < \omega \leq \pi)$

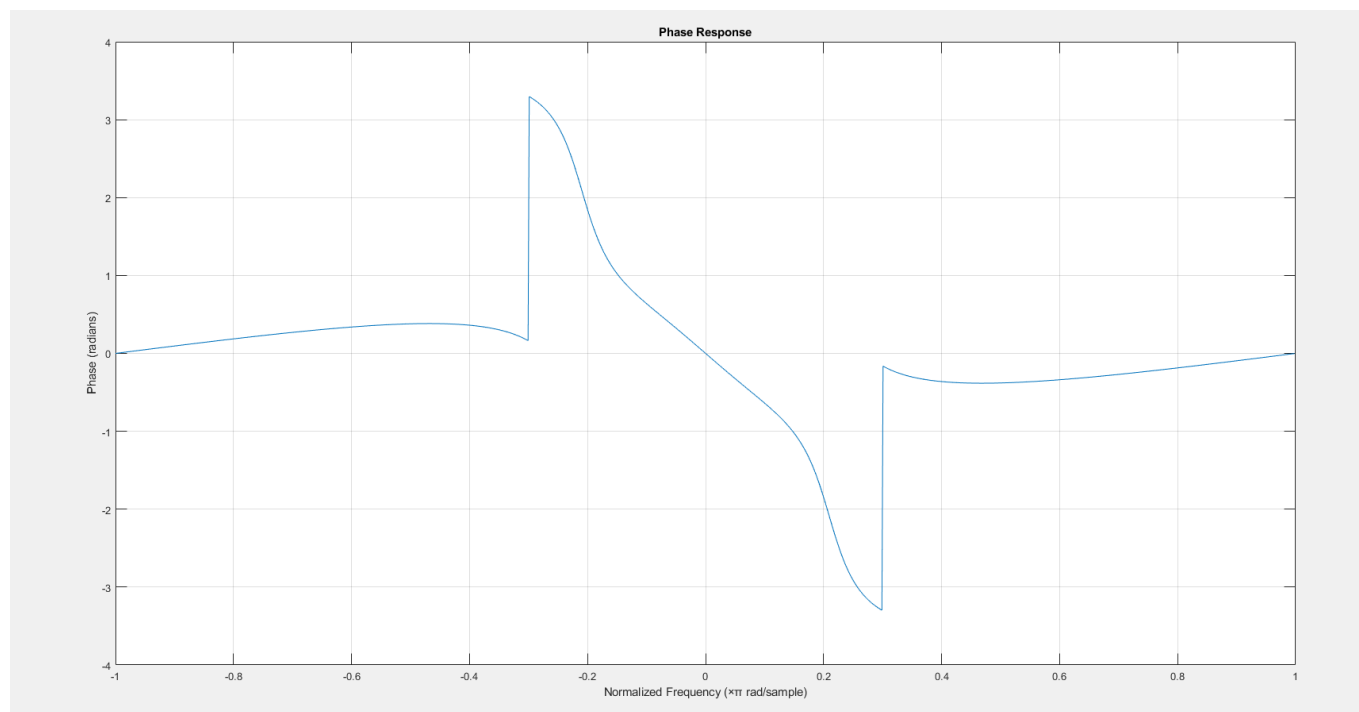


Magnitude response of the filter in dB in the frequency range $(-w_p < \omega \leq w_p)$

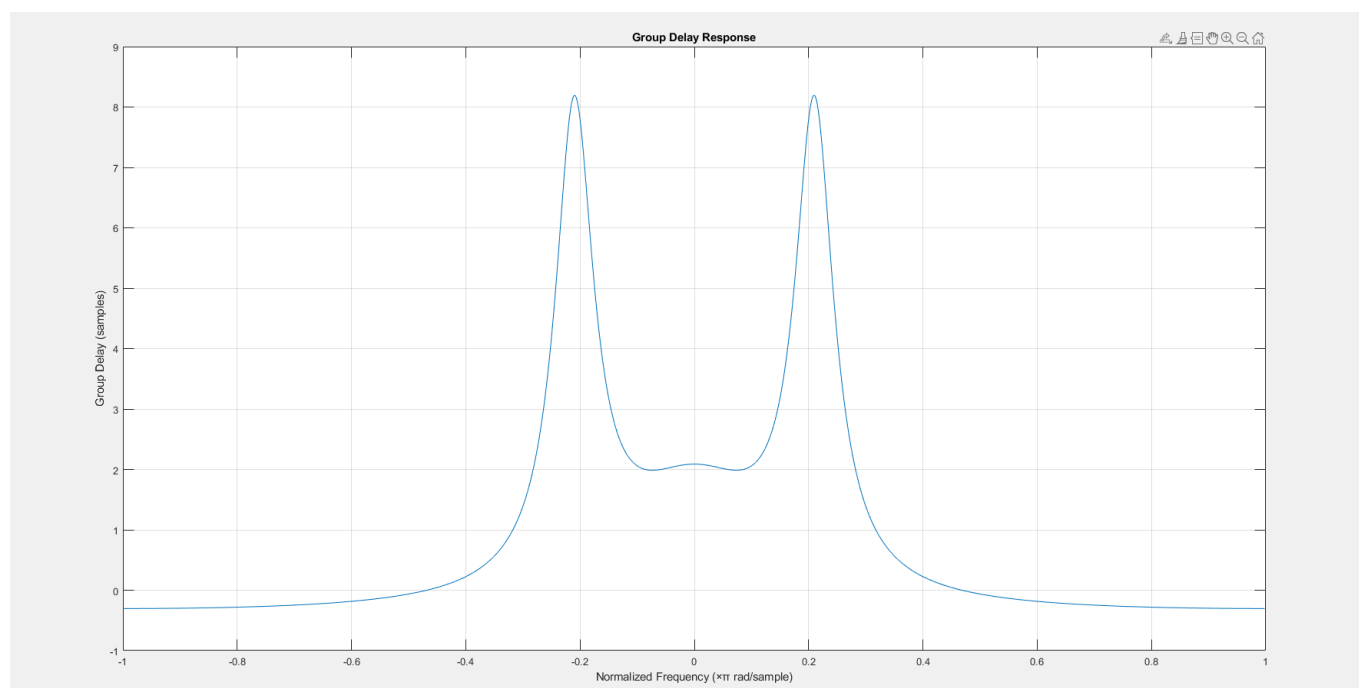


From the graph, the passband ripples $A_p \approx 0.53$ dB

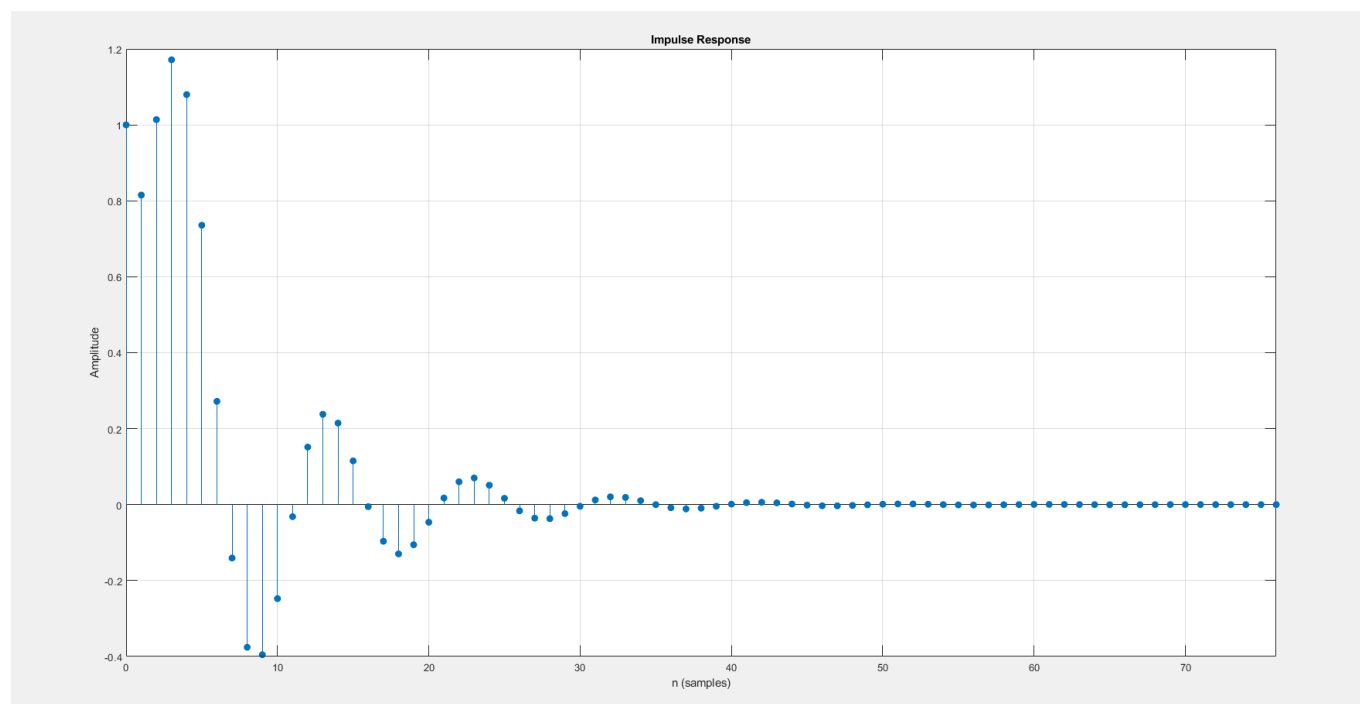
Phase response of the filter in the frequency range $(-\pi < \omega \leq \pi)$ (Using unwrap function)



Group delay of the filter in the frequency range $(-\pi < \omega \leq \pi)$

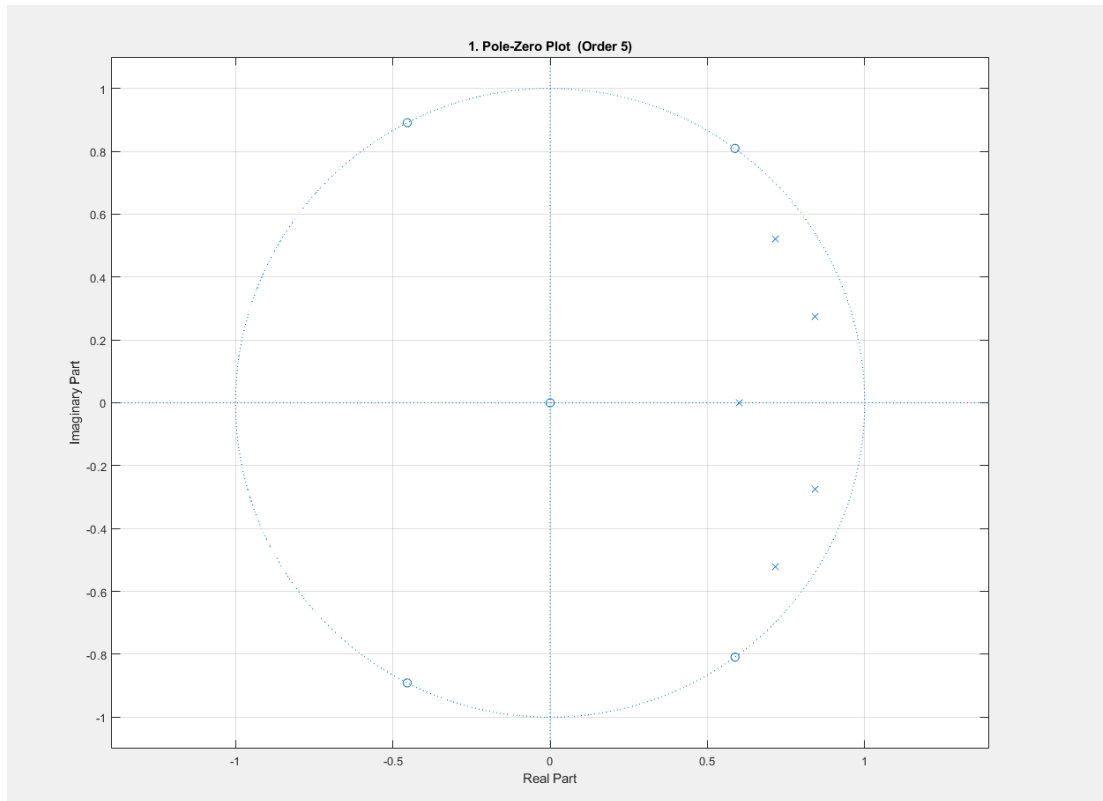


The filter's impulse response

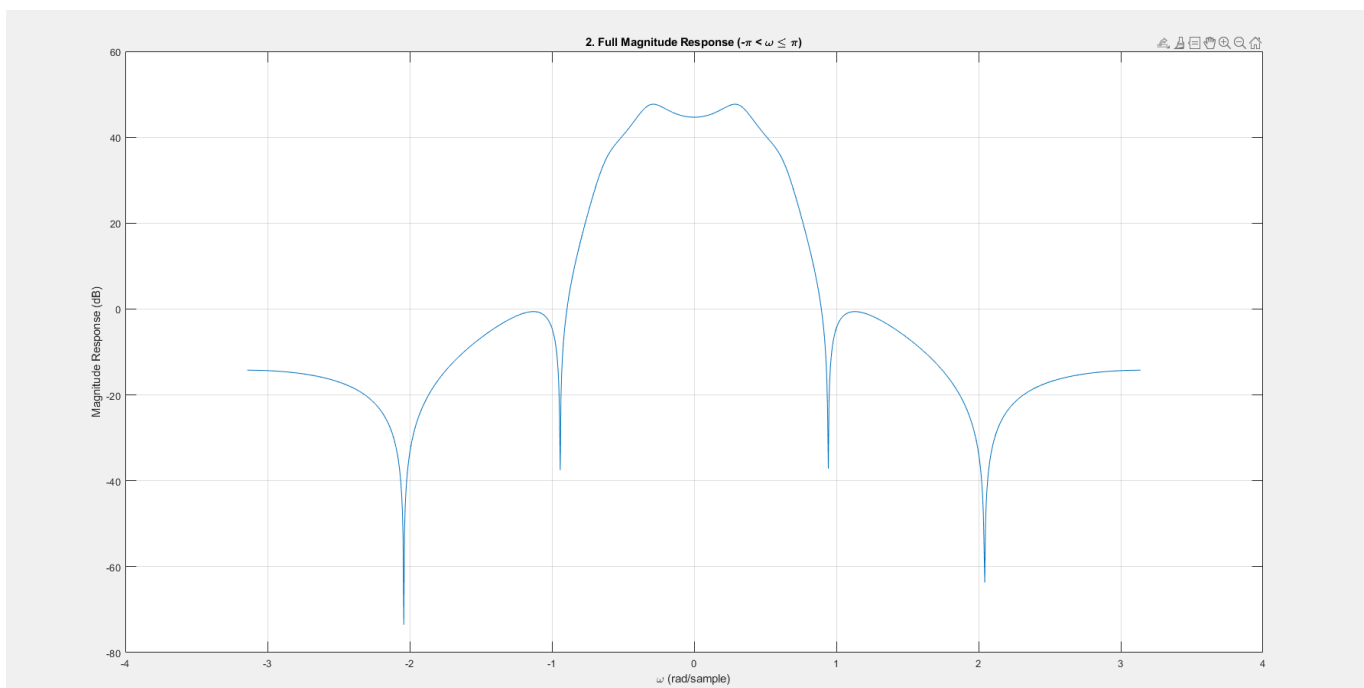


Fifth order discrete-time filter

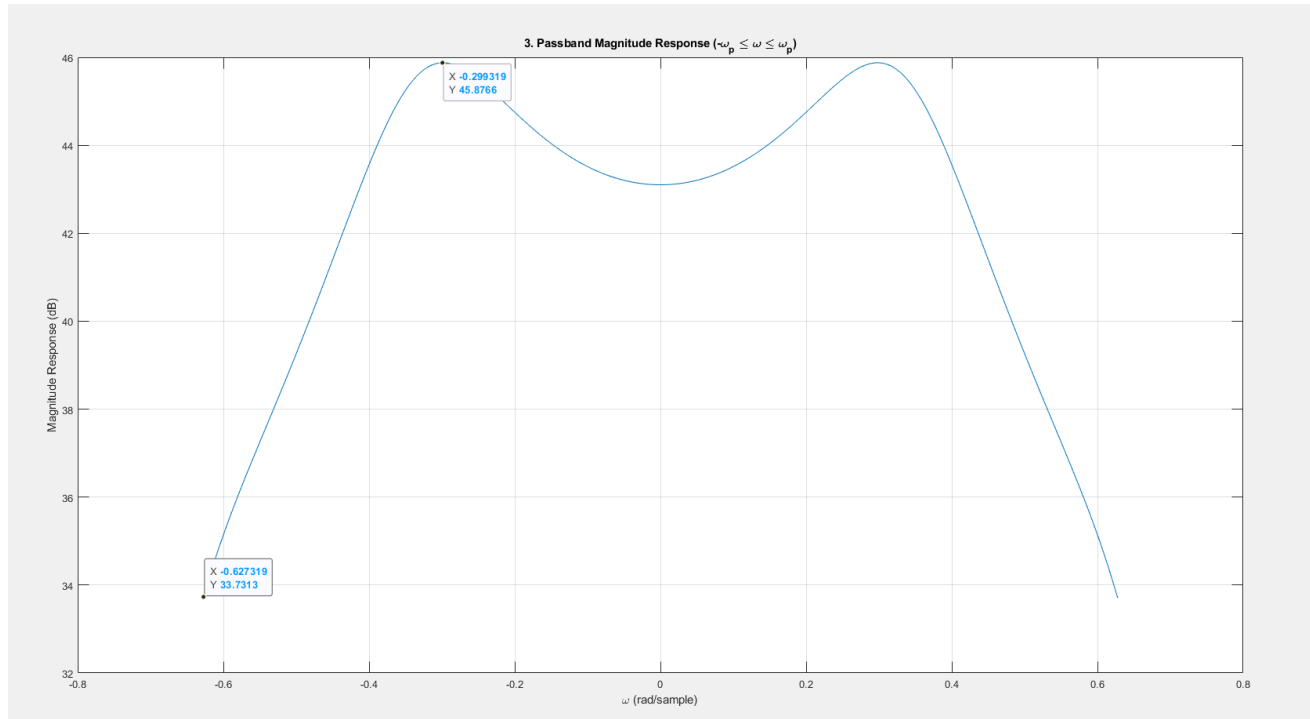
Pole-zero plot and the unit circle in the Z-domain



Magnitude response of the filter in dB in the frequency range $(-\pi < \omega \leq \pi)$

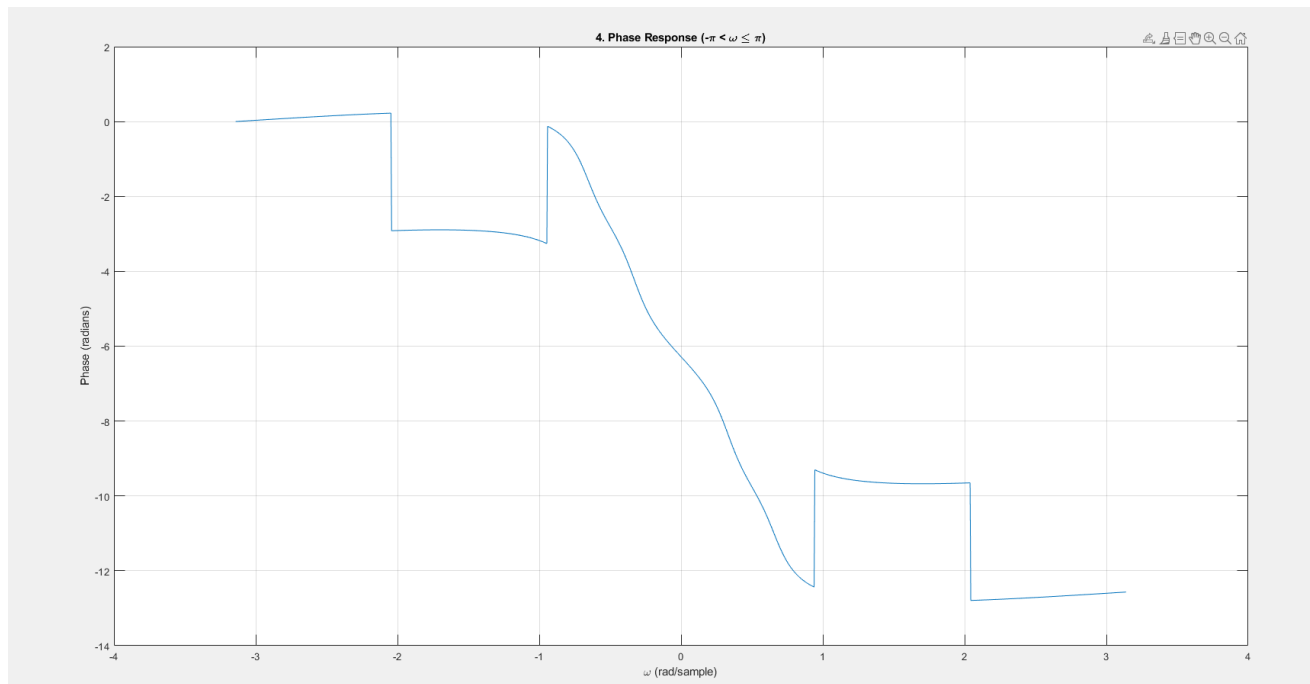


Magnitude response of the filter in dB in the frequency range $(-w_p < w \leq w_p)$

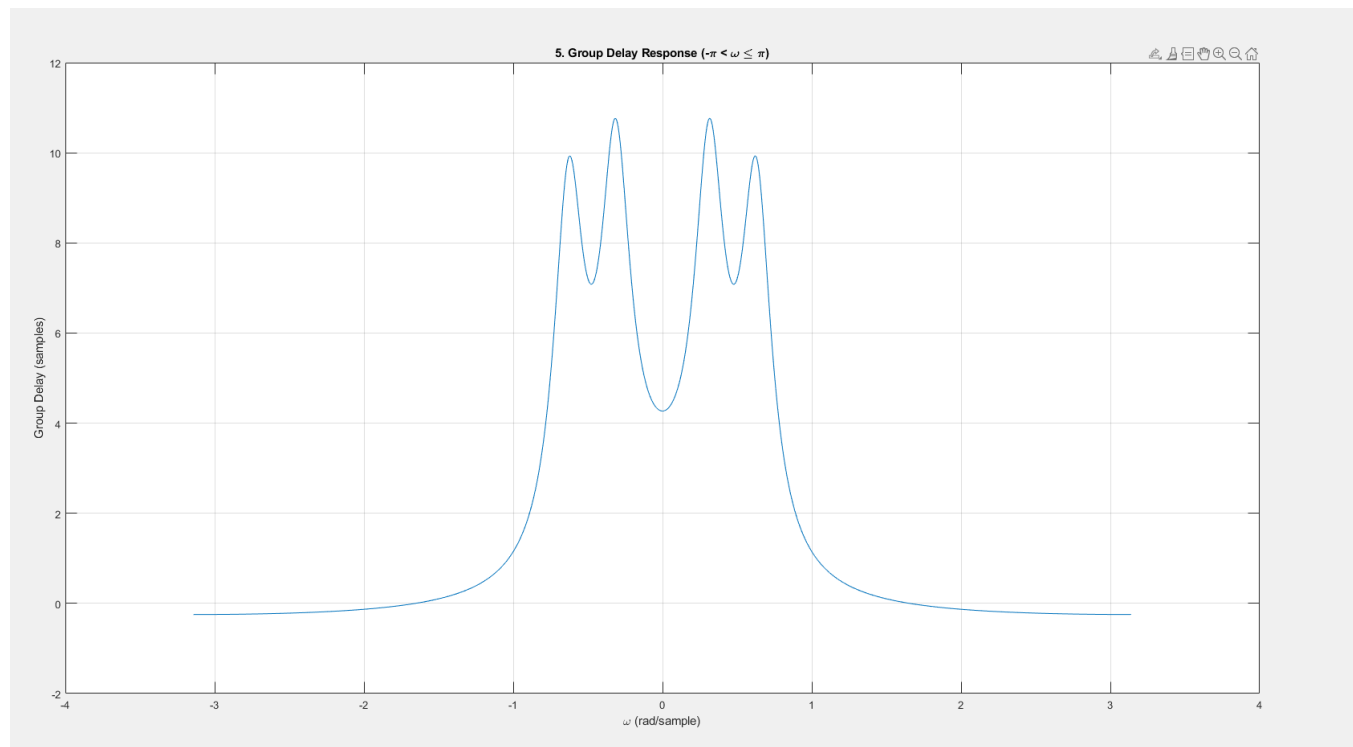


The passband ripples from the graph = $45.8766 - 33.7313 = 12.1453$ dB

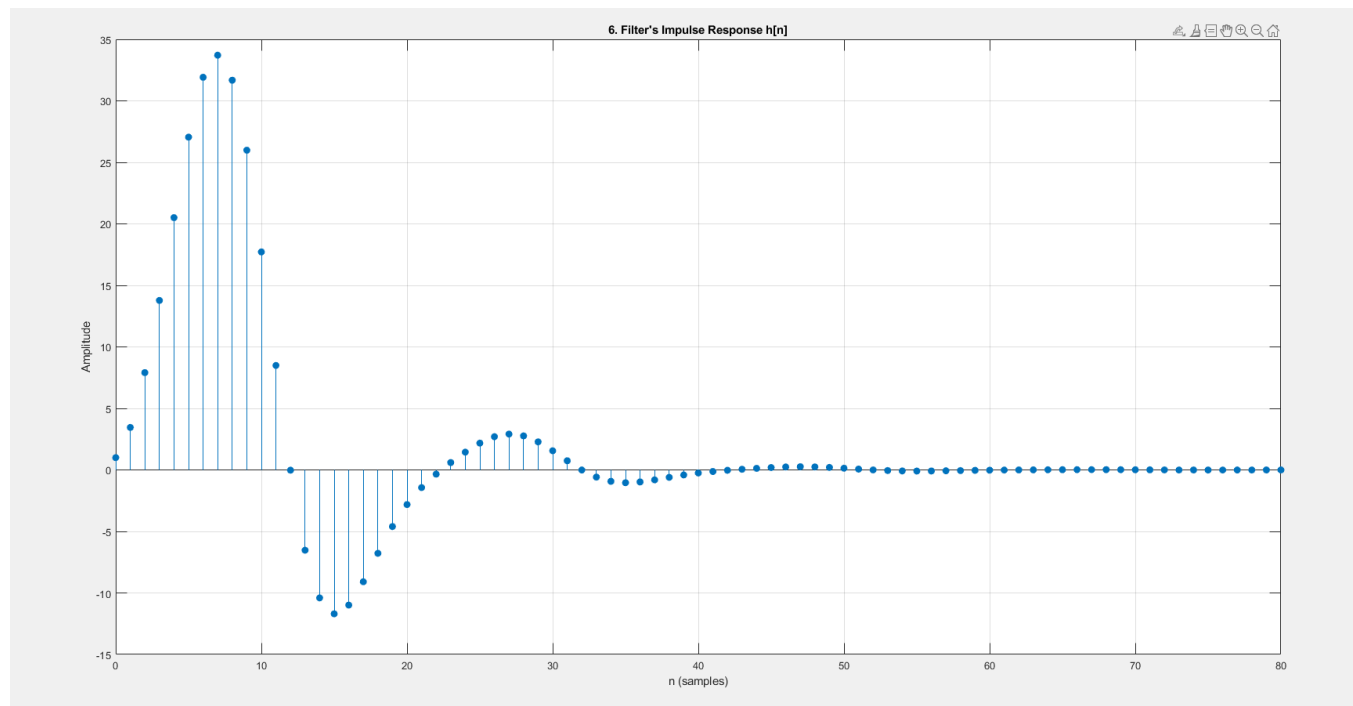
Phase response of the filter in the frequency range $(-\pi < w \leq \pi)$ (Using unwrap function)



Group delay of the filter in the frequency range $(-\pi < \omega \leq \pi)$



The filter's impulse response



A snippet from the Code showing the Sweeping on r1, r2 and r3 to get passband ripples less than 0.5 dB

```

338 %% Sweep on r1 , r2 and r3
339
340 r1_values = 0.1 : 0.01 : 0.99;
341 r2_values = 0.1 : 0.01 : 0.99;
342 r3_values = 0.1 : 0.01 : 0.99;
343
344 Optimum_r1 = 0;
345 Optimum_r2 = 0;
346 Optimum_r3 = 0;
347 Optimum_ripple = inf;
348
349 for r1_trial = r1_values
350
351     for r2_trial = r2_values
352
353         for r3_trial = r3_values
354
355             % Complex Zeros
356             z1 = exp(1j*wso);
357             z2 = exp(-1j*wso);
358             w_z_order5 = (wso + pi)/2;
359             z3 = exp(1j*w_z_order5);
360             z4 = exp(-1j*w_z_order5);
361
362             zeros_vec = [z1; z2; z3; z4];
363
364             % Poles
365             p1 = r2_trial;
366             p2 = r1_trial * exp(1j*optimum_wp1);
367             p3 = r1_trial * exp(-1j*optimum_wp1);
368             w_p_order5 = (0 + optimum_wp1)/2;
369             p4 = r3_trial * exp(1j*w_p_order5);
370             p5 = r3_trial * exp(-1j*w_p_order5);
371
372             poles_vec = [p1; p2; p3; p4; p5];
373
374             % Transfer Function Coefficients
375             b = real(poly(zeros_vec));
376             a = real(poly(poles_vec));
377
378             % Frequency Response in Passband
379             w_passband = linspace(0, wpo, 2000);
380
381             H_trial = freqz(b, a, w_passband);
382
383             % Normalize Gain
384             H_trial = H_trial / abs(H_trial(1));
385
386             % Magnitude in dB
387             Hdb = 20*log10(abs(H_trial));
388
389             % Passband Ripple
390             ripple = max(Hdb) - min(Hdb);
391
392         end
393     end
394 end

```

Where wso (stopband edge) = 0.3π , and $Optimum_wp1 = 0.21\pi$ from third order sweep

```
392         % Save Optimum Parameters
393
394         if ripple < Optimum_ripple
395
396             Optimum_ripple = ripple;
397
398             Optimum_r1 = r1_trial;
399             Optimum_r2 = r2_trial;
400             Optimum_r3= r3_trial;
401         end
402
403     end
404 end
405 end
406
407 % Optimum Parameters
408
409 fprintf('==== Optimum Parameters ==== \n');
410
411 fprintf('Optimum r1  = %.3f \n', Optimum_r1);
412 fprintf('Optimum r2  = %.3f \n', Optimum_r2);
413 fprintf('Optimum r3  = %.3f \n', Optimum_r3);
414 fprintf('Passband Ripple = %.3f dB \n', Optimum_ripple);
415
```

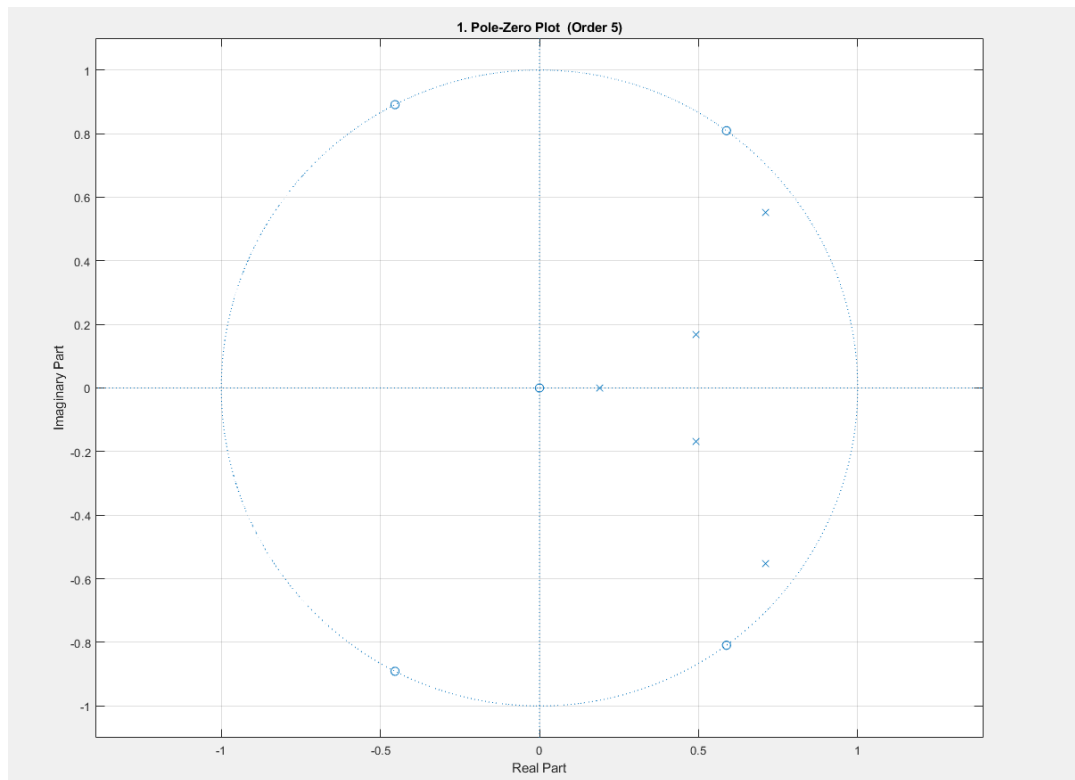
Command Window

```
==== Optimum Parameters ====
Optimum r1  = 0.900
Optimum r2  = 0.190
Optimum r3  = 0.520
Passband Ripple = 0.444 dB
fx >> |
```

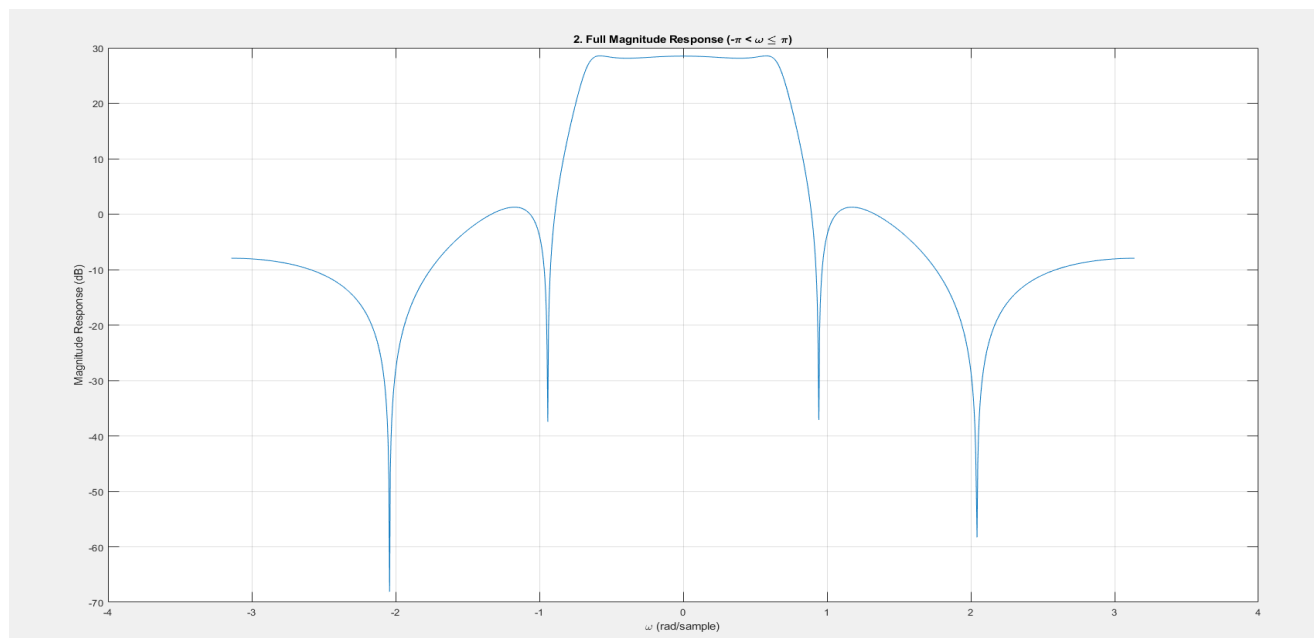
Note: Run section of this snippet code in MATLAB file to see the results

The new Plots for the optimum r_1 , r_2 and r_3 and passband ripples = 0.444 dB

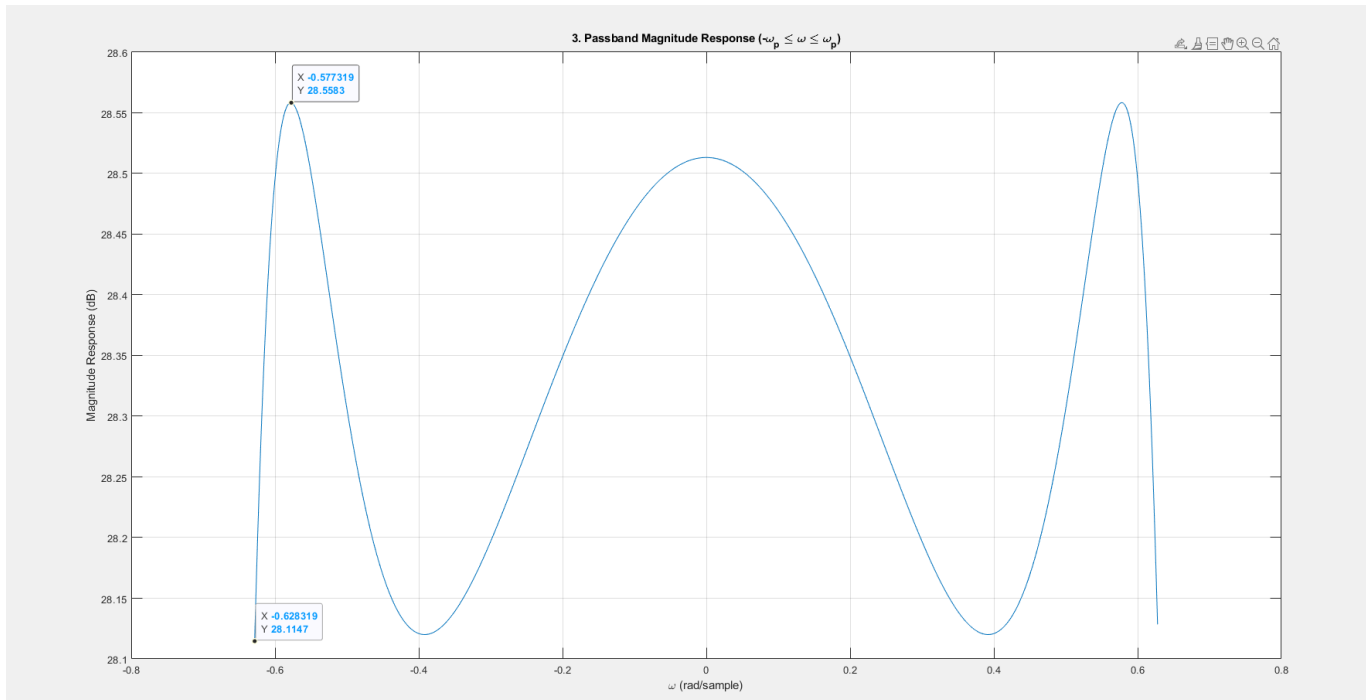
Pole-zero plot and the unit circle in the Z-domain



Magnitude response of the filter in dB in the frequency range $(-\pi < \omega \leq \pi)$

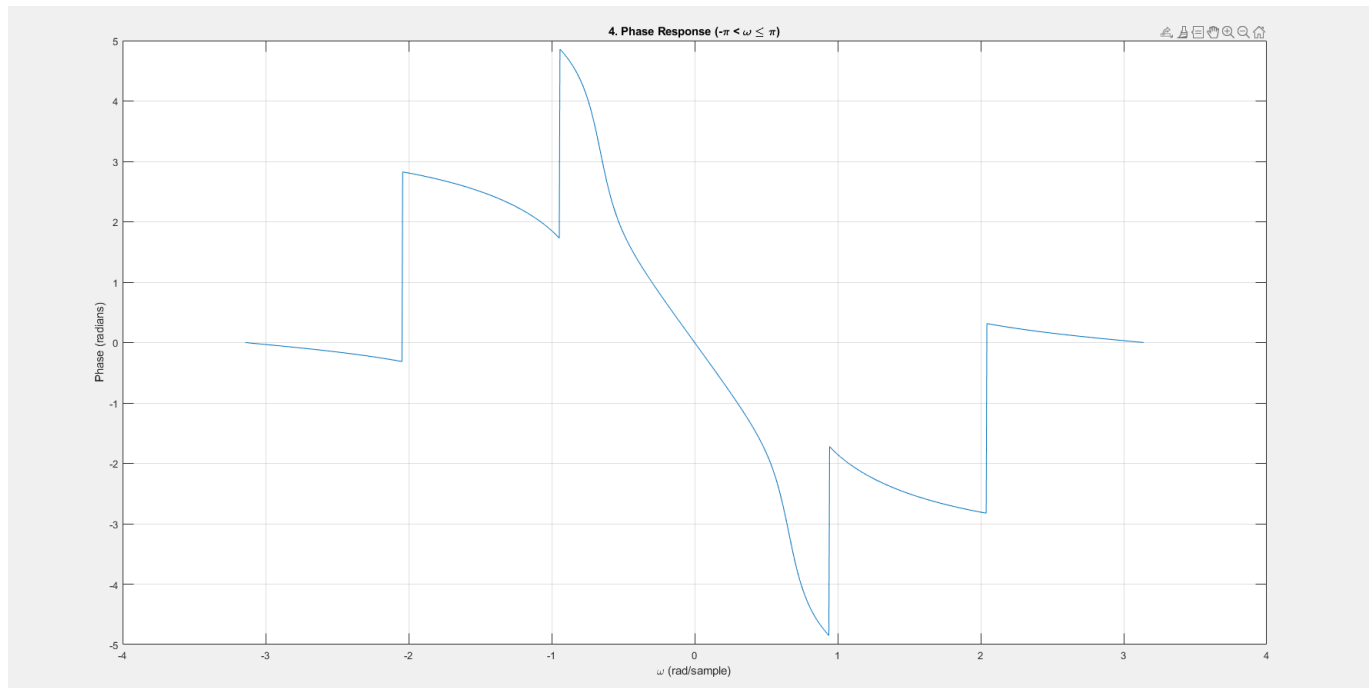


Magnitude response of the filter in dB in the frequency range $(-w_p < w \leq w_p)$

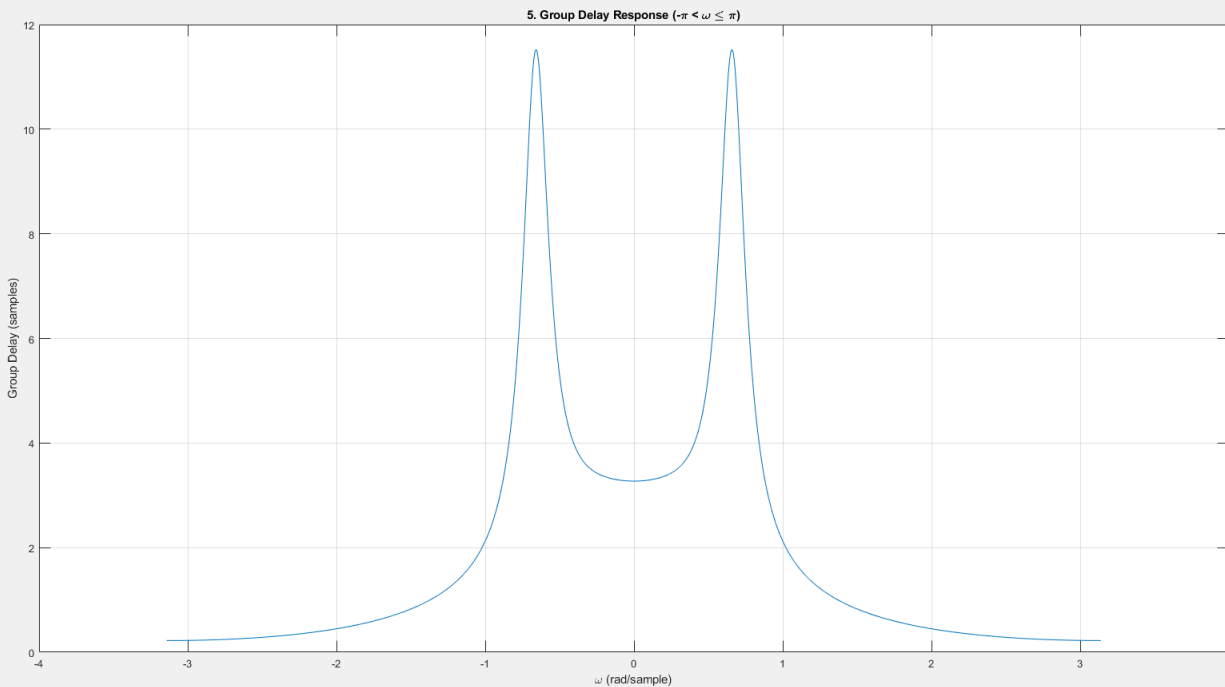


From the graph, $A_p = 28.5583 - 28.1147 = \mathbf{0.4436 \text{ dB}}$

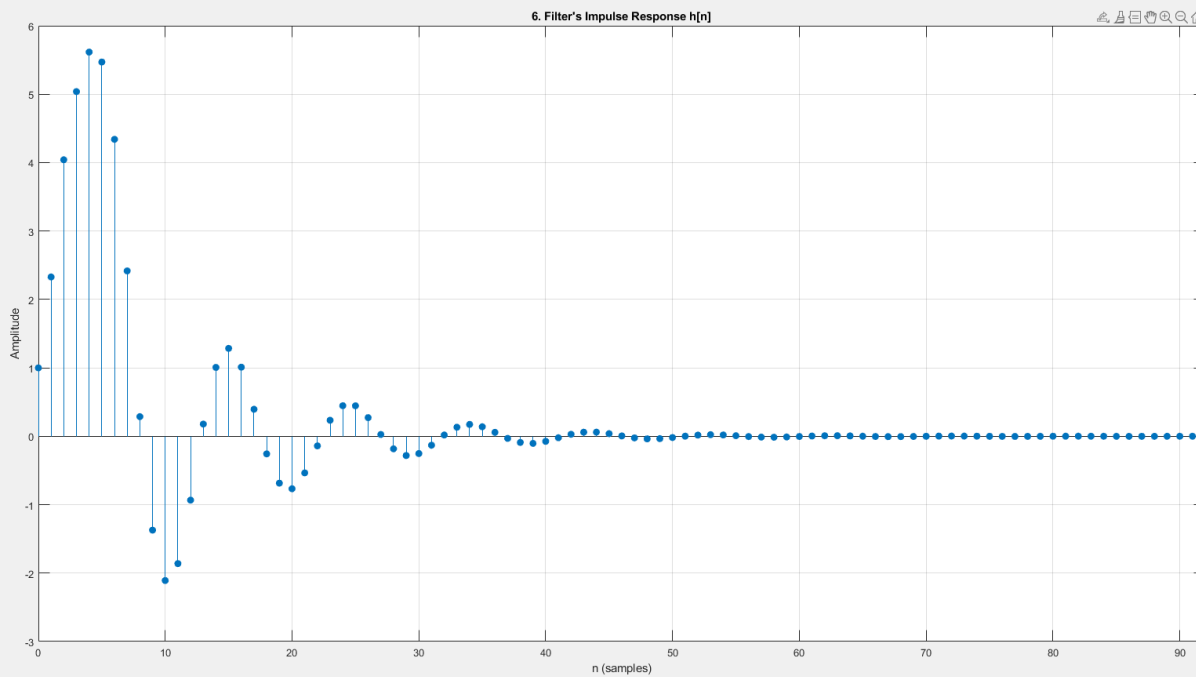
Phase response of the filter in the frequency range $(-\pi < w \leq \pi)$ (Using unwrap function)



Group delay of the filter in the frequency range $(-\pi < \omega \leq \pi)$



The filter's impulse response



Frequency Transformation using Pole-Zero Pattern Rotation (HPF)

Note: The transformation was made for the fifth order LPF after the adjustment of the magnitude of poles to obtain passband ripple less than 0.5 dB

For a digital LPF $\rightarrow$ HPF transformation, use:

$$H_{\text{HPF}}(Z) = H_{\text{LPF}}(-Z)$$

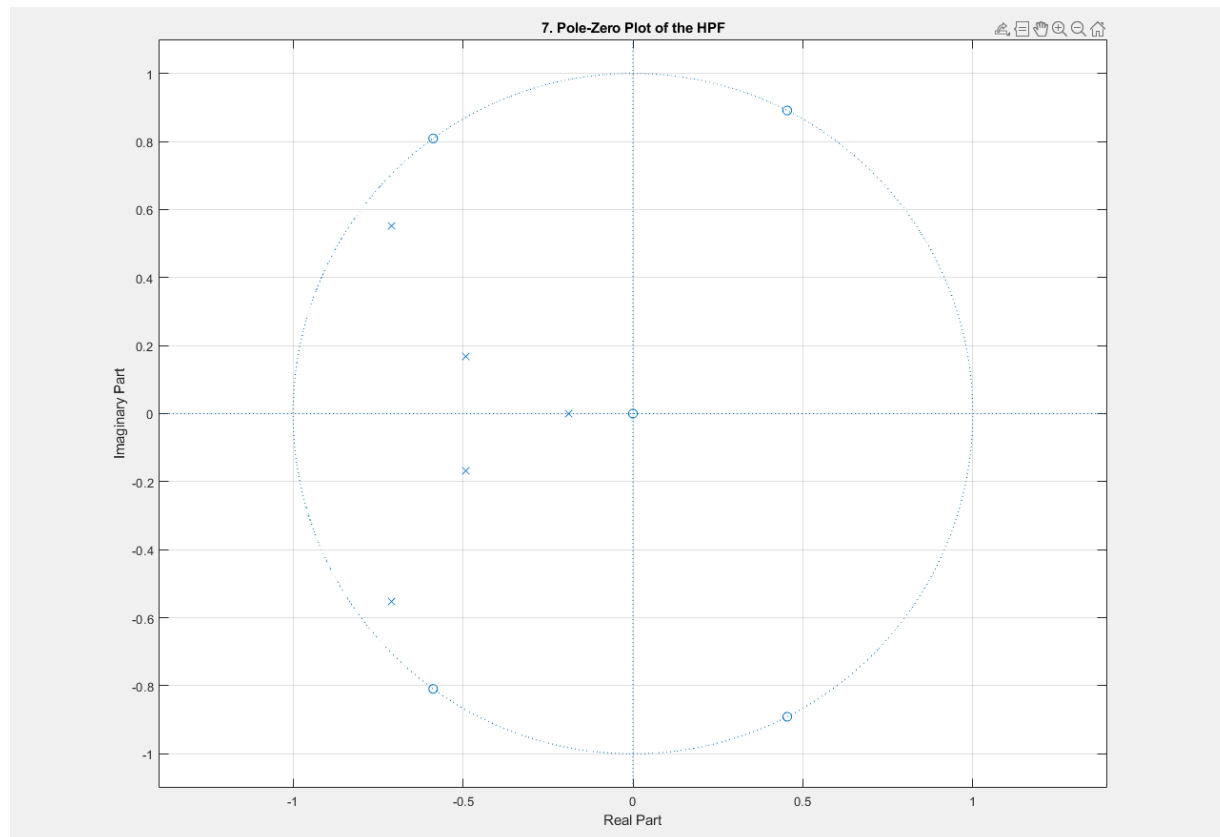
This means replacing z^{-1} by $-z^{-1}$

```

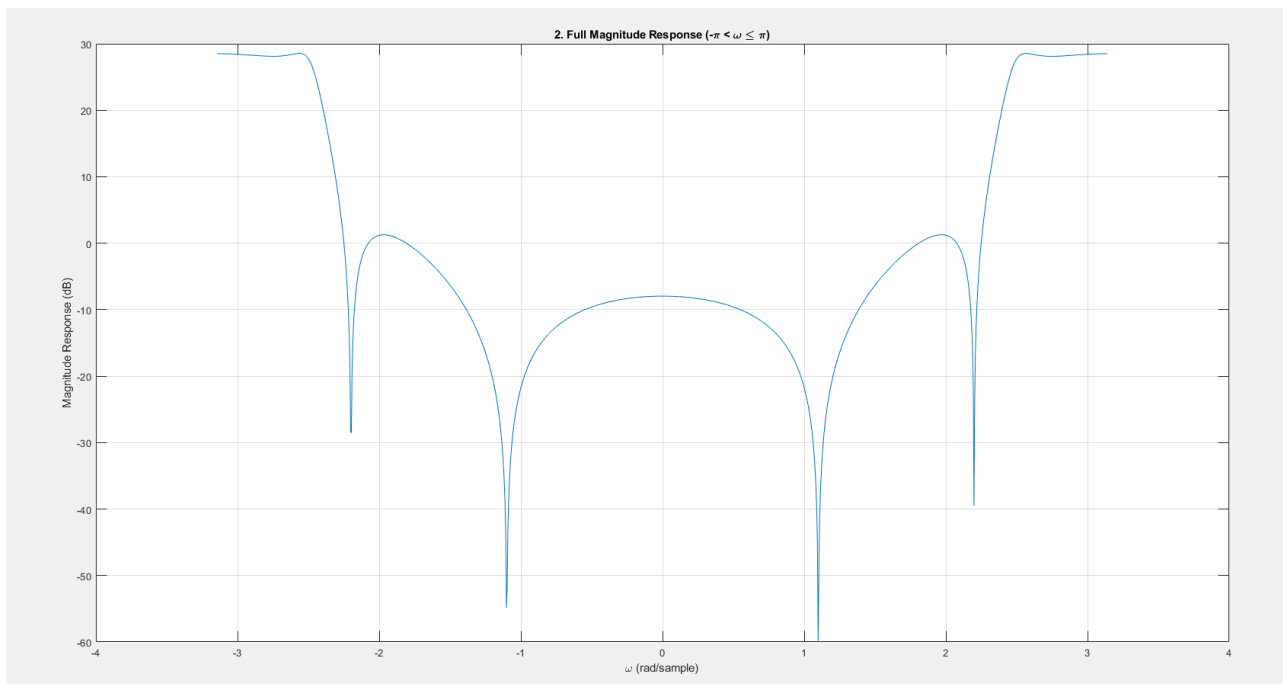
531  %% Frequency Transformation using Pole-Zero Pattern Rotation (HPF)
532
533  zeros_HPF = -zeros_vec;          % Rotate all zeros by 180° on the z-plane
534  poles_HPF = -poles_vec;         % Rotate all poles by 180° on the z-plane
535
536  b_HPF = poly(zeros_HPF);
537  a_HPF = poly(poles_HPF);
538

```

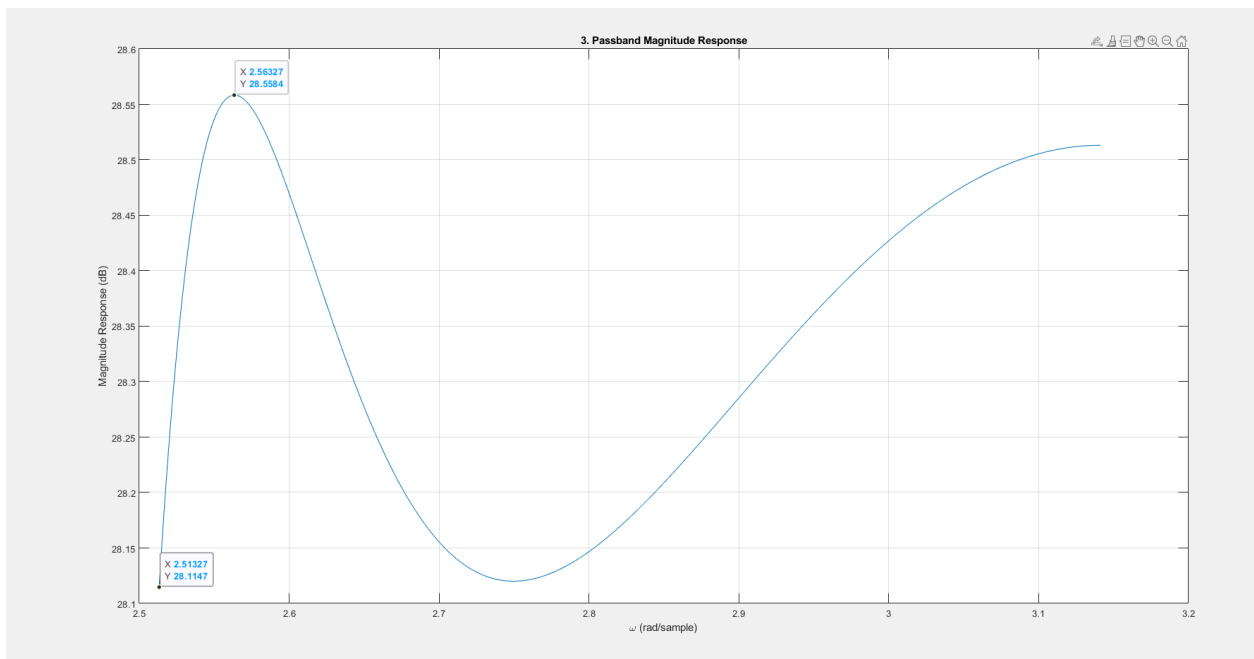
Pole-zero plot and the unit circle in the Z-domain



Magnitude response of the filter in dB in the frequency range $(-\pi < \omega \leq \pi)$

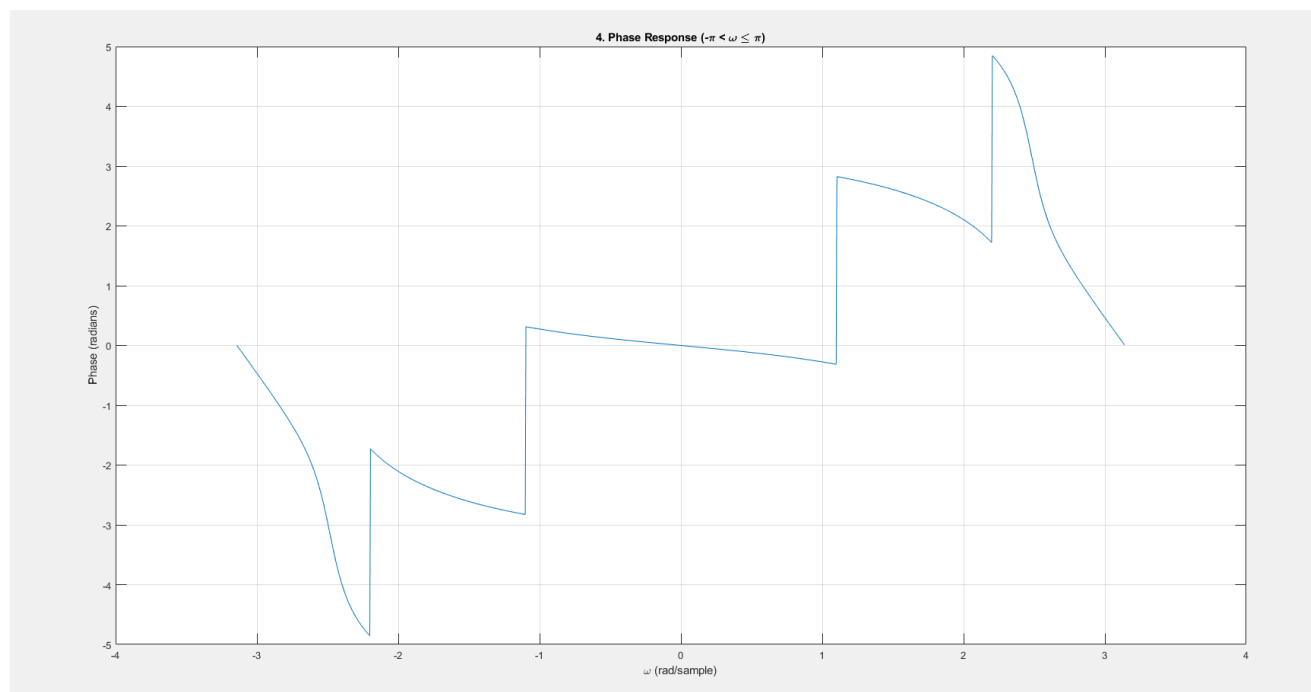


Magnitude response of the filter in dB in passband region $(\pi - \omega_p < \omega \leq \pi)$

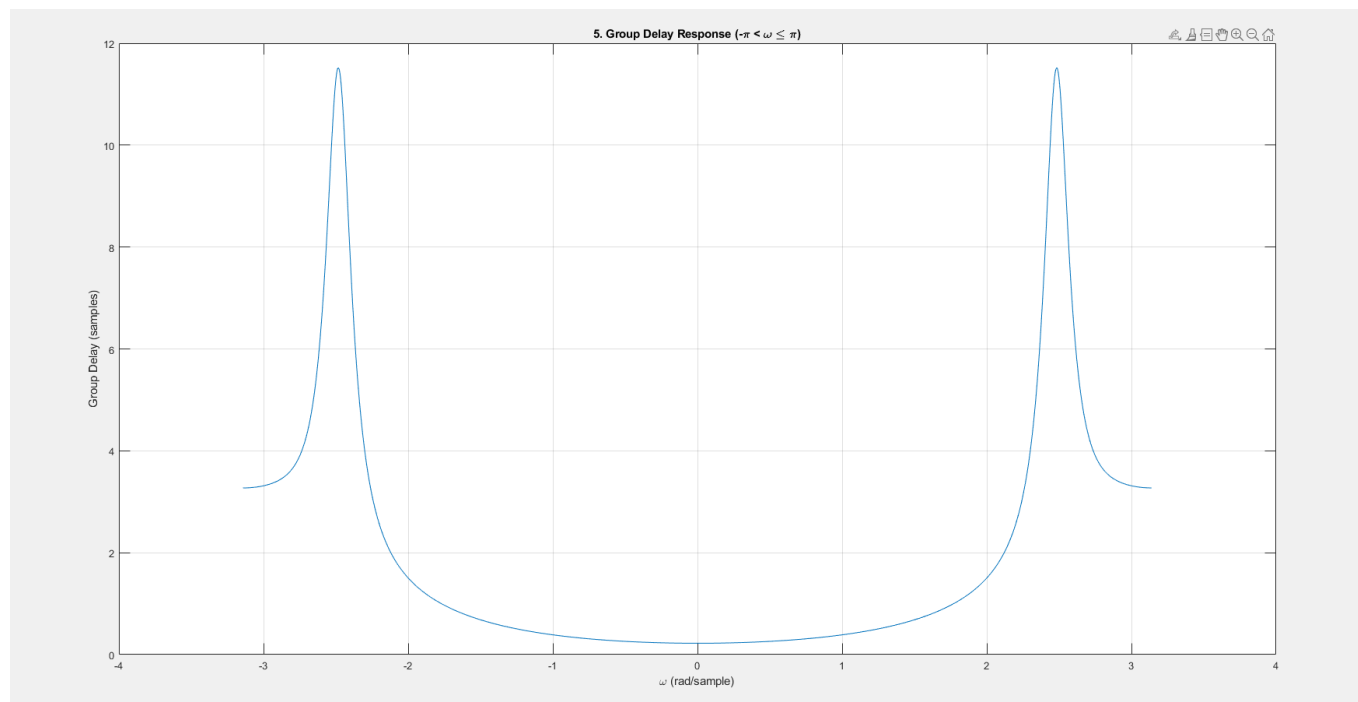


$$A_p \approx 0.4437 \text{ dB}$$

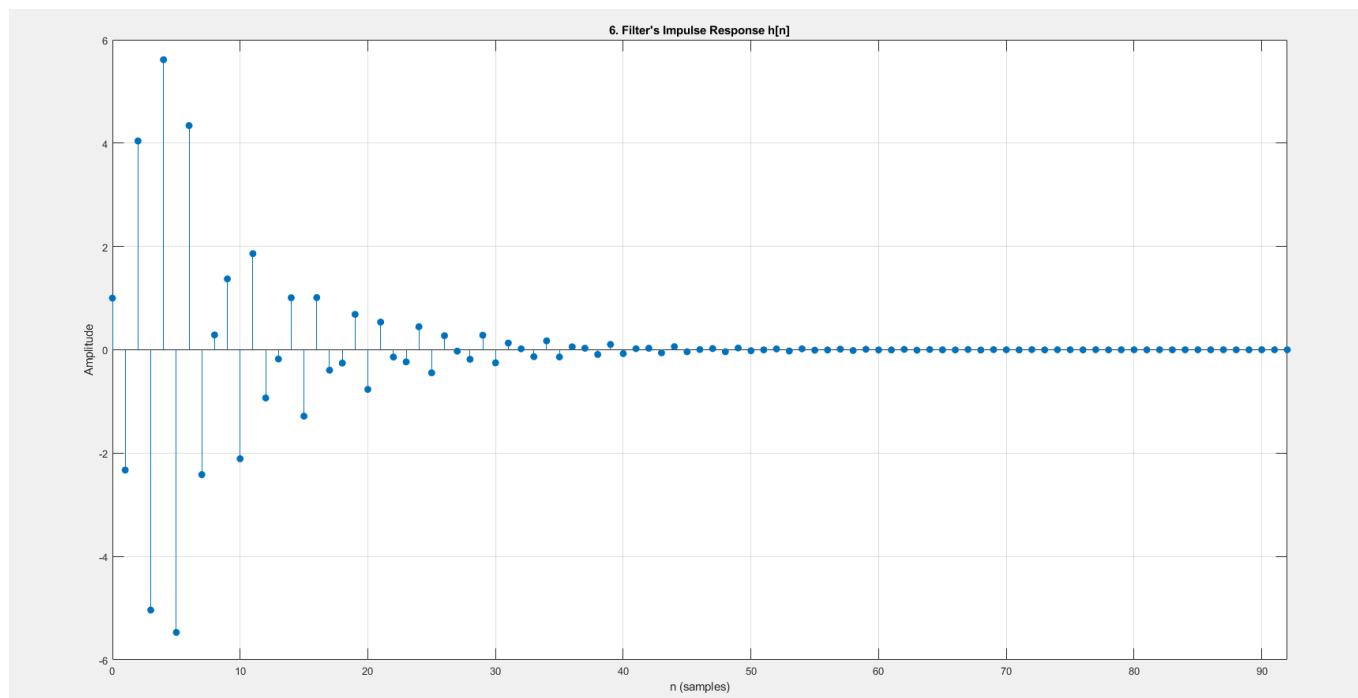
Phase response of the filter in the frequency range $(-\pi < \omega \leq \pi)$ (Using unwrap function)



Group delay of the filter in the frequency range $(-\pi < \omega \leq \pi)$



The filter's impulse response



Frequency Transformation using Pole-Zero Pattern Rotation (BPF)

To ensure that the BPF coefficients are real-valued, all poles and zeros are placed in complex conjugate pairs. Since the polynomial coefficients are formed from conjugate root pairs, the resulting numerator and denominator coefficients are real. Any small imaginary parts caused by numerical round-off are removed using the real function in MATLAB.

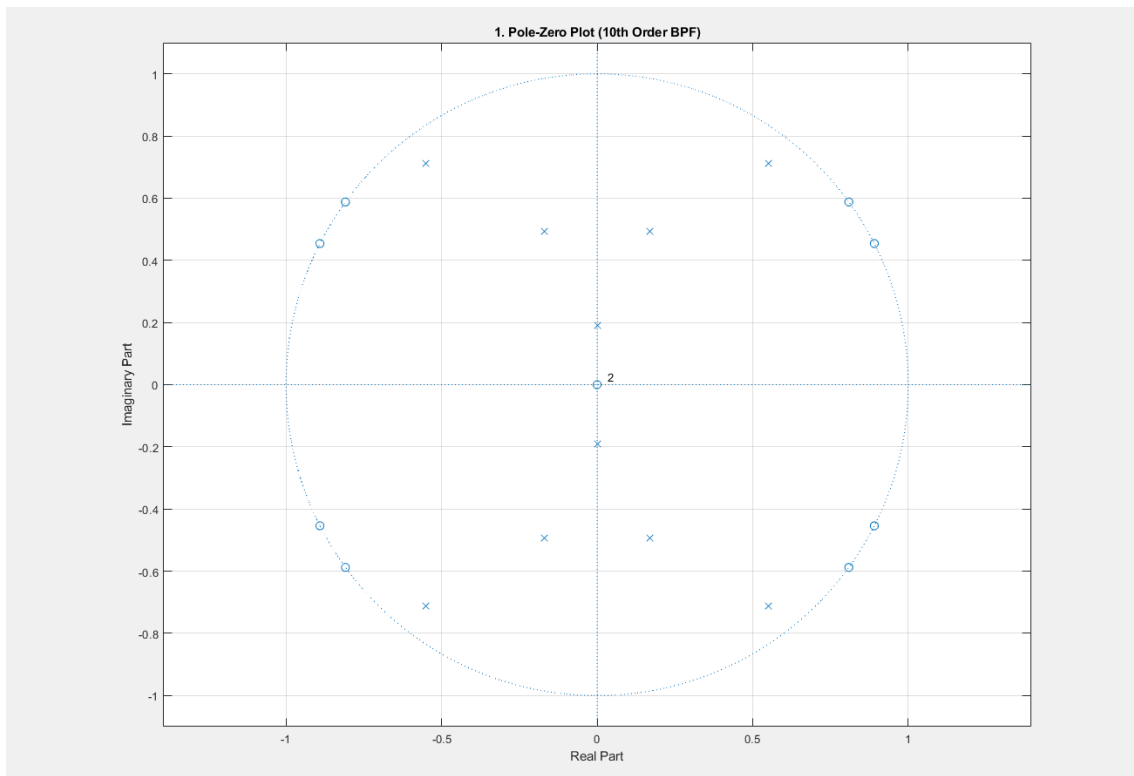
we must include both $+\pi/2$ and $-\pi/2$ shifts. This doubles the order of the filter (5th order LPF becomes 10th order BPF).

```

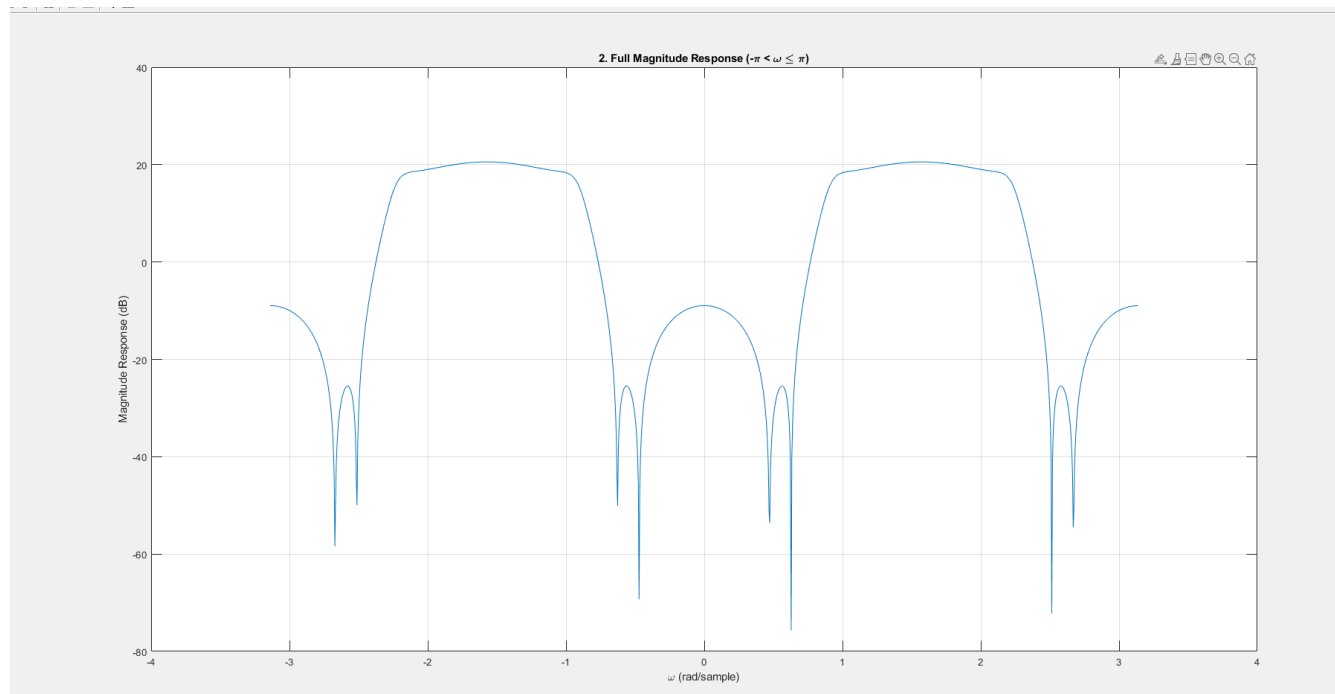
36 shift_pos = exp(1j * pi/2);
37 shift_neg = exp(-1j * pi/2);
38
39 zeros_vec_bpf = [zeros_vec_lpf * shift_pos; zeros_vec_lpf * shift_neg];
40 poles_vec_bpf = [poles_vec_lpf * shift_pos; poles_vec_lpf * shift_neg];
41
42 % Filter BPF Coefficients |
43 b_bpf = real(poly(zeros_vec_bpf));
44 a_bpf = real(poly(poles_vec_bpf));
45

```

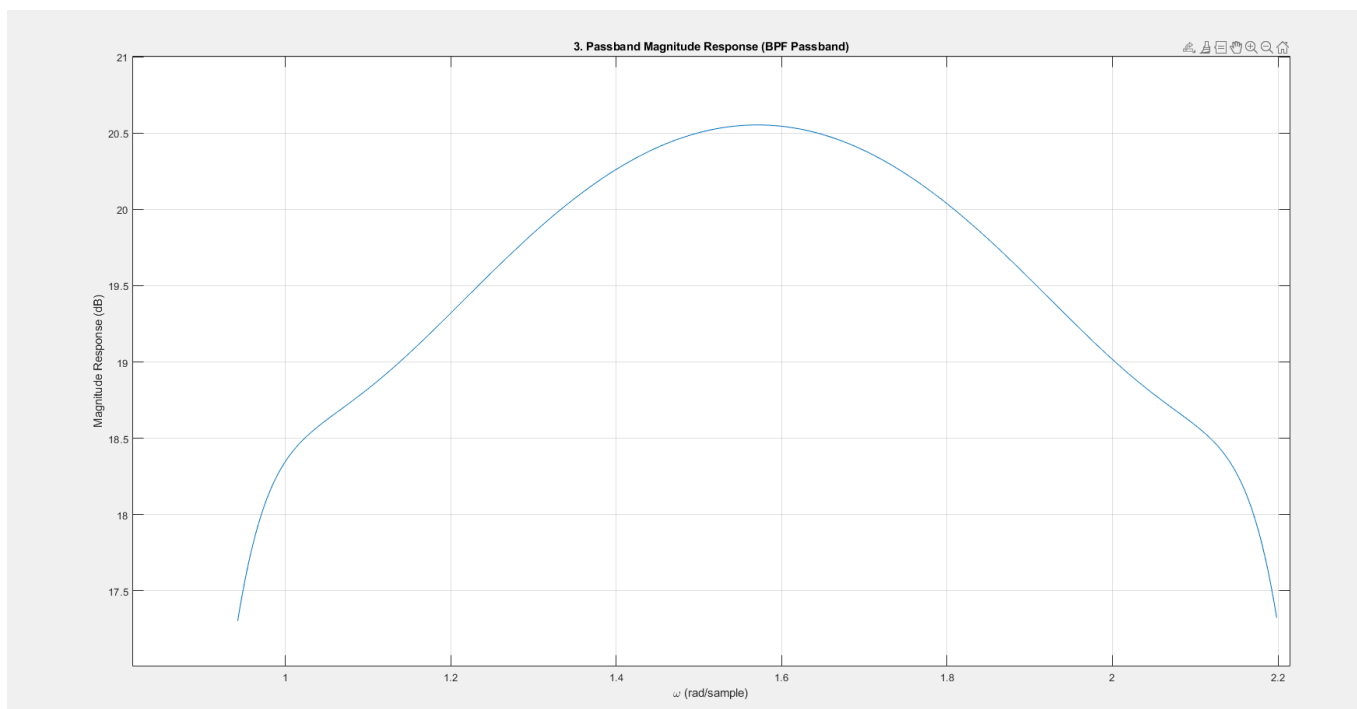
Pole-zero plot and the unit circle in the Z-domain



Magnitude response of the filter in dB in the frequency range $(-\pi < \omega \leq \pi)$

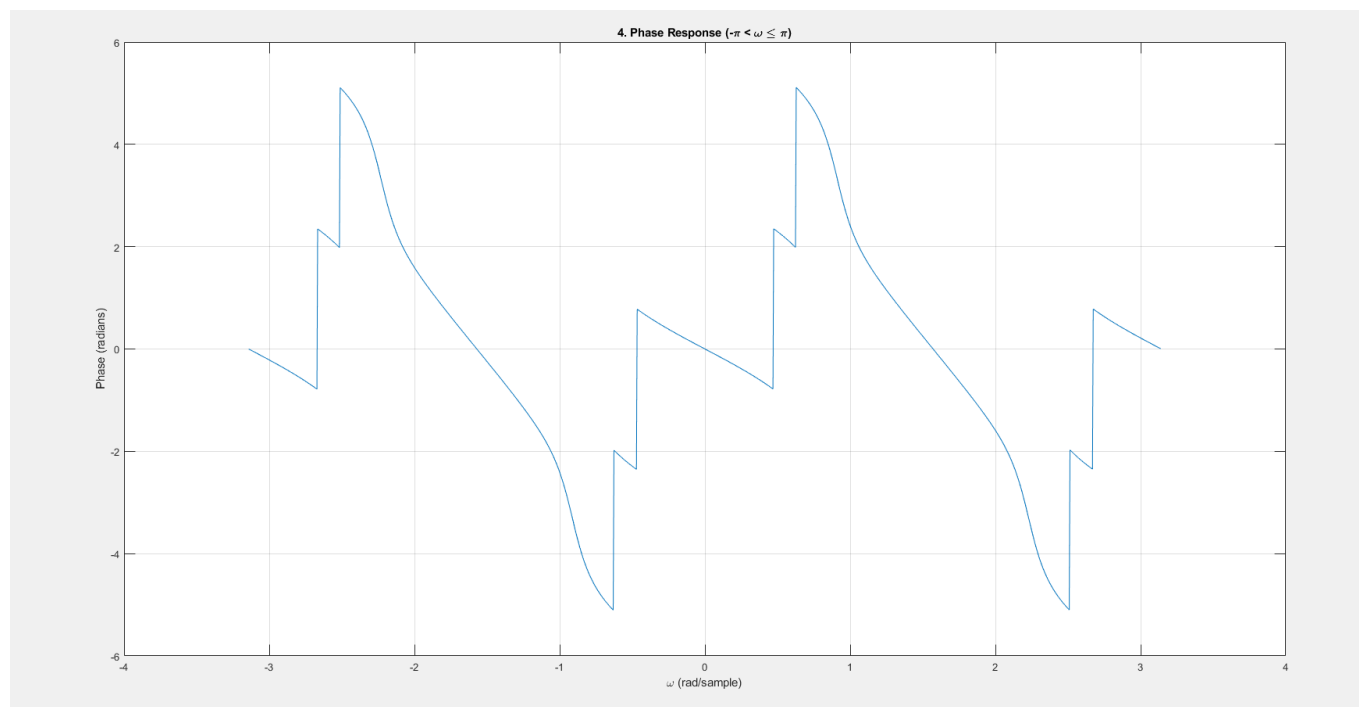


Magnitude response of the filter in dB in passband region $(\frac{\pi}{2} - \omega_p < \omega \leq \frac{\pi}{2} + \omega_p)$

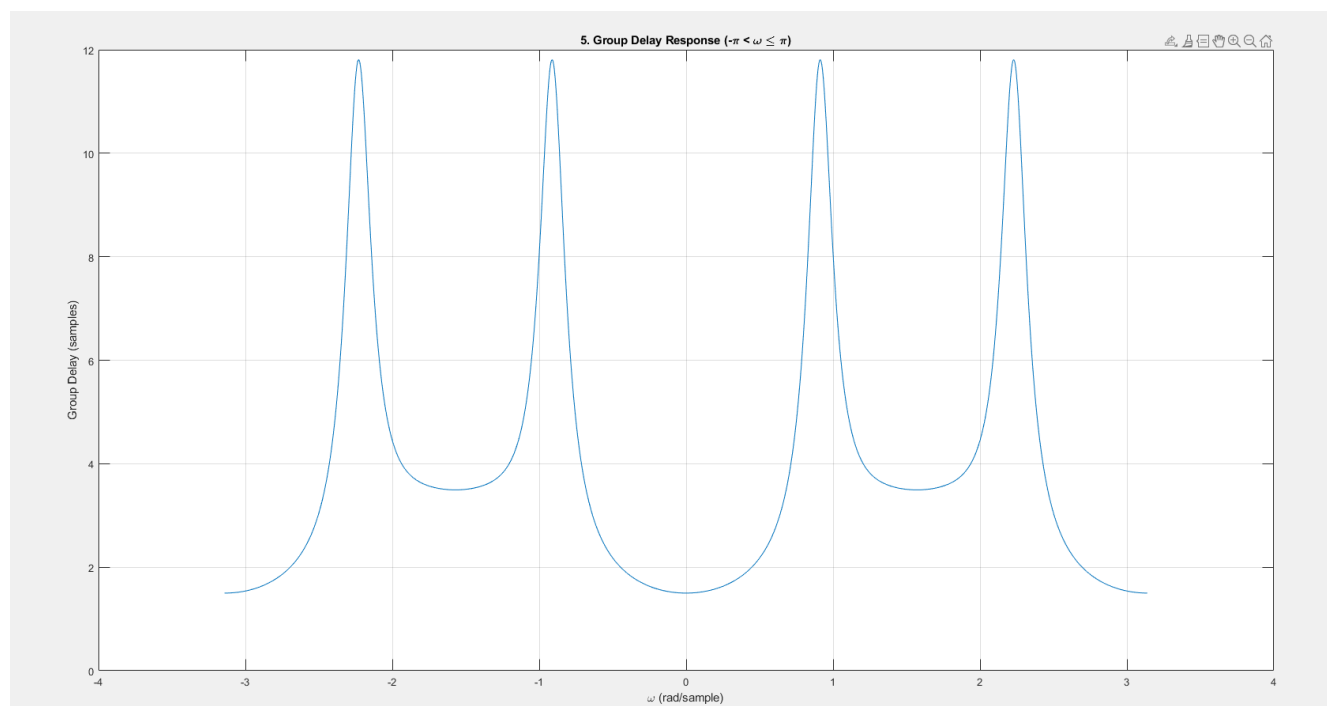


Passband Ripples (A_p) ≈ 3.25 dB

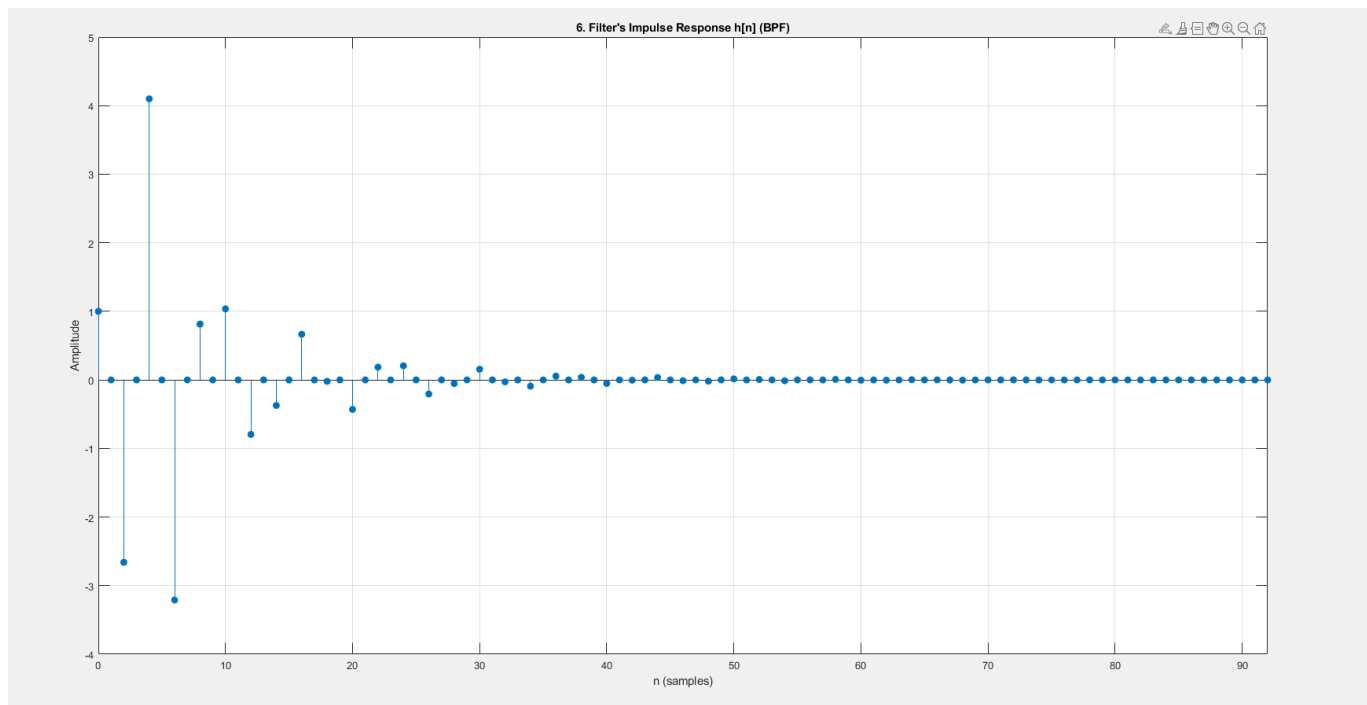
Phase response of the filter in the frequency range $(-\pi < \omega \leq \pi)$ (Using unwrap function)



Group delay of the filter in the frequency range $(-\pi < \omega \leq \pi)$



The filter's impulse response



Comb Filter

Hand Analysis

Hand analysis:

$$H_{\text{comb}}(z) = \sum_{n=-\infty}^{\infty} h_{\text{comb}}[n] z^{-n}$$

$$H_{\text{comb}}(z) = \sum_{n=-\infty}^{\infty} \left(\sum_{m=-\infty}^{\infty} h[m] \delta[n-m] \right) z^{-n}$$

$$H_{\text{comb}}(z) = \sum_{m=-\infty}^{\infty} h[m] \sum_{n=-\infty}^{\infty} \delta[n-m] z^{-n}$$

$$H_{\text{comb}}(z) = \sum_{m=-\infty}^{\infty} h[m] z^{-m} = \sum_{m=-\infty}^{\infty} h[m] (z^{-1})^{-m}$$

$$\boxed{H_{\text{comb}}(z) = H(z^{-1})}$$

$$H_{\text{comb}}(e^{j\omega}) = H(e^{j\omega L})$$

$$\therefore H_{\text{comb}}(e^{j(\omega + \frac{2\pi}{L})}) = H(e^{jL(\omega + \frac{2\pi}{L})})$$

$$\therefore H_{\text{comb}}(e^{j(\omega + \frac{2\pi}{L})}) = H(e^{jL\omega} e^{j2\pi})$$

$$\boxed{\therefore H_{\text{comb}}(e^{j(\omega + \frac{2\pi}{L})}) = H(e^{jL\omega}) = H_{\text{comb}}(e^{j\omega})}$$

$\therefore H_{\text{comb}}(e^{j\omega})$ is periodic with period $\frac{2\pi}{L}$ radians

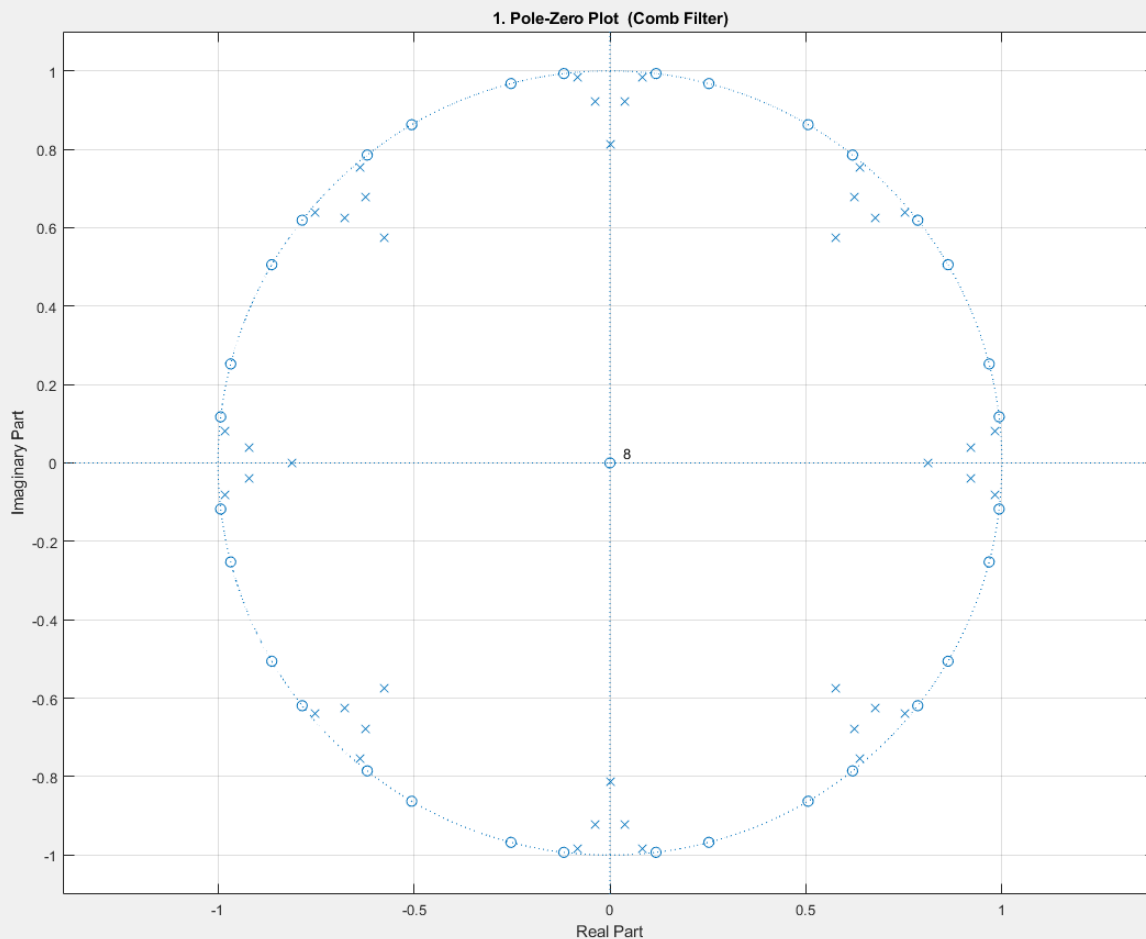
A comb filter with $L = 8$ was obtained from the fifth-order LPF by replacing z^{-1} with z^{-8} , i.e., $H_{\text{comb}}(z) = H(z^8)$. This operation inserts seven zeros between every pair of coefficients in both the numerator and denominator polynomials, resulting in a frequency response that repeats every $\frac{2\pi}{8}$

```

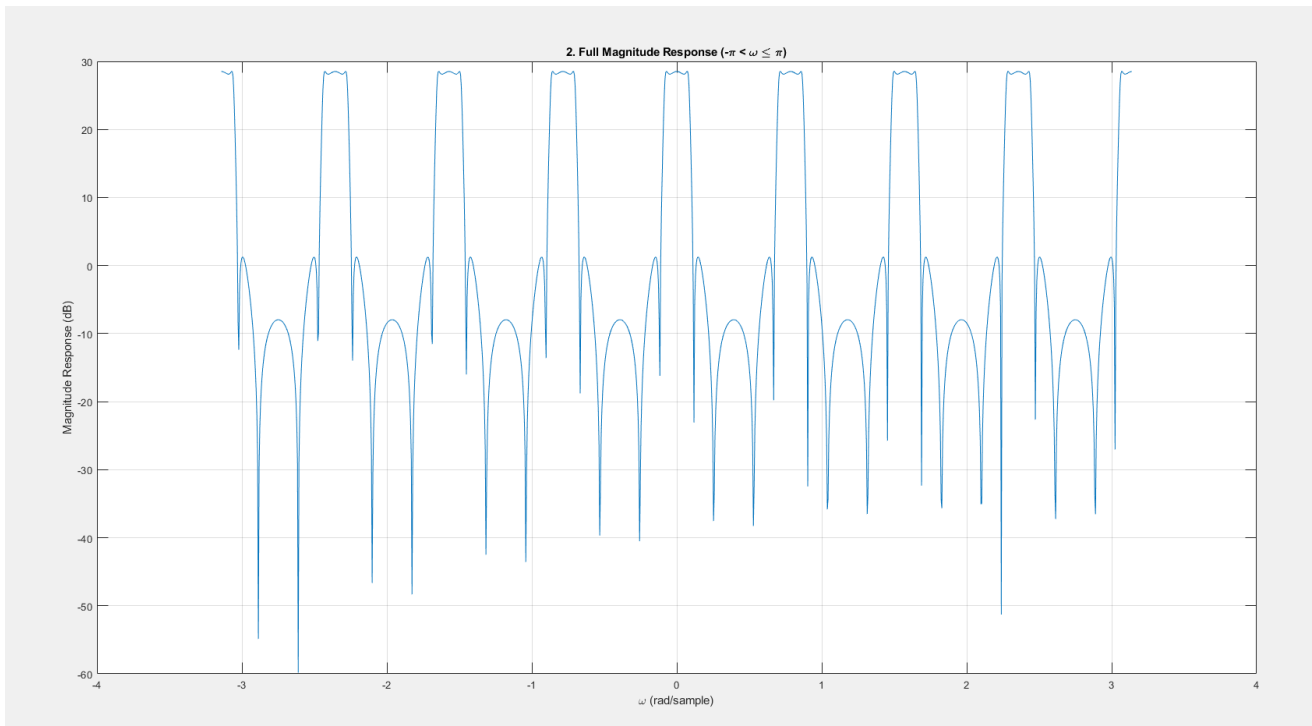
543 %% Filter Coefficients
544 b = poly(zeros_vec); % Numerator coefficients
545 a = poly(poles_vec); % Denominator coefficients
546
547 L = 8; % Comb filter factor
548
549 b_comb = zeros(1, (length(b)-1)*L + 1); % New numerator coefficients
550 a_comb = zeros(1, (length(a)-1)*L + 1); % New denominator coefficients
551
552 b_comb(1:L:end) = b; % Insert zeros between numerator coefficients
553 a_comb(1:L:end) = a; % Insert zeros between denominator coefficients
554

```

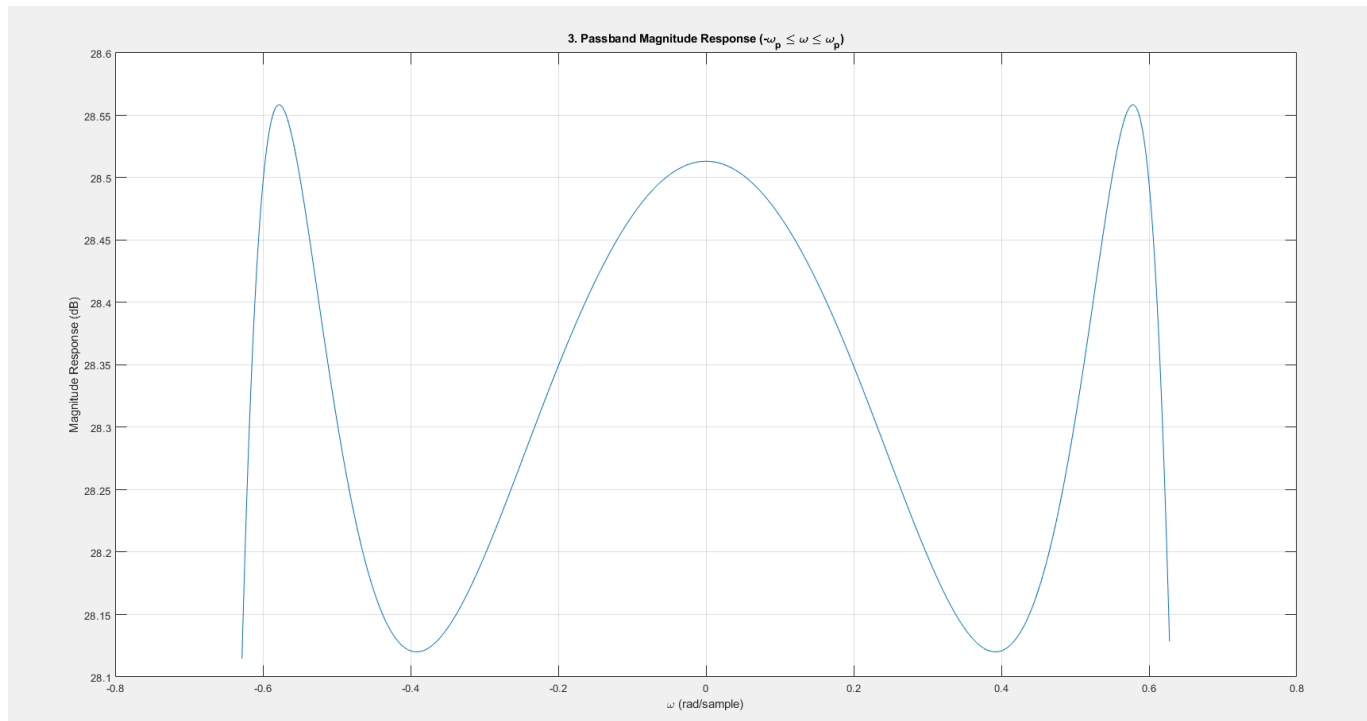
Pole-zero plot and the unit circle in the Z-domain



Magnitude response of the filter in dB in the frequency range $(-\pi < \omega \leq \pi)$

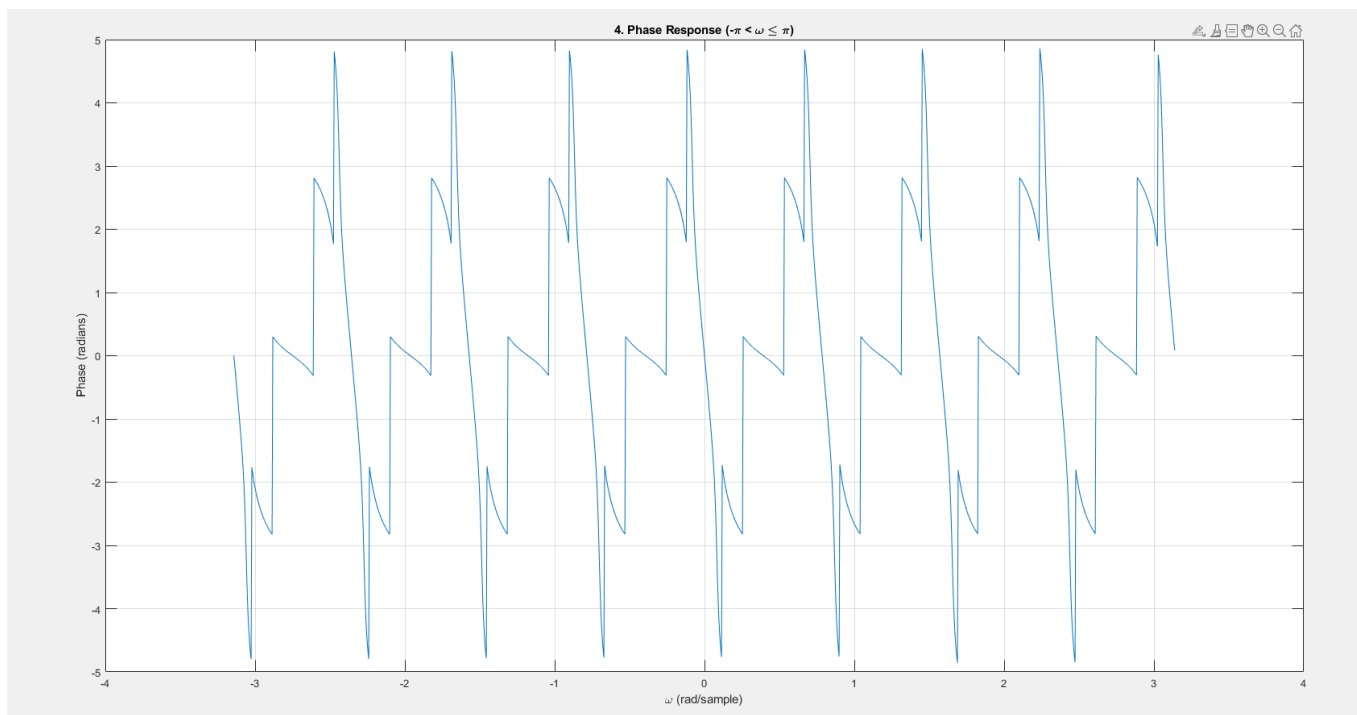


Magnitude response of the filter in dB in the frequency range $(-\omega_p < \omega \leq \omega_p)$

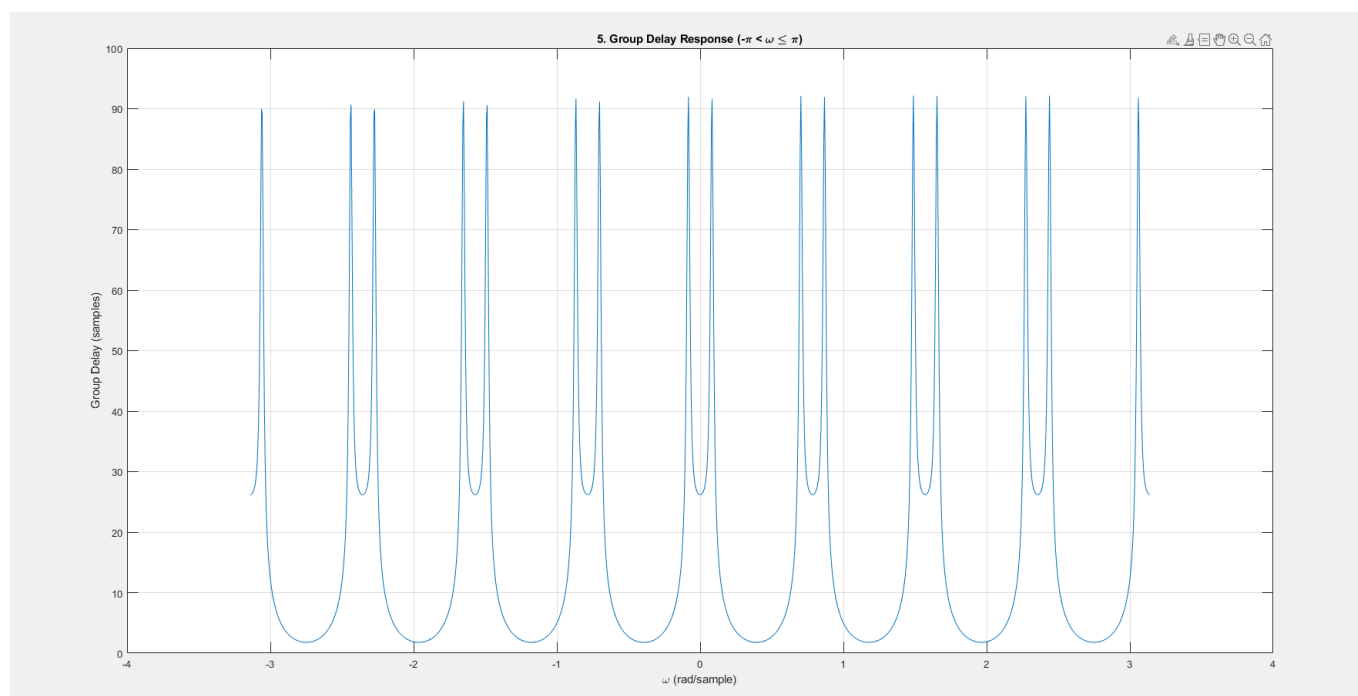


$A_p \approx 0.4436 \text{ dB}$ (Same as the calculated one in fifth order filter)

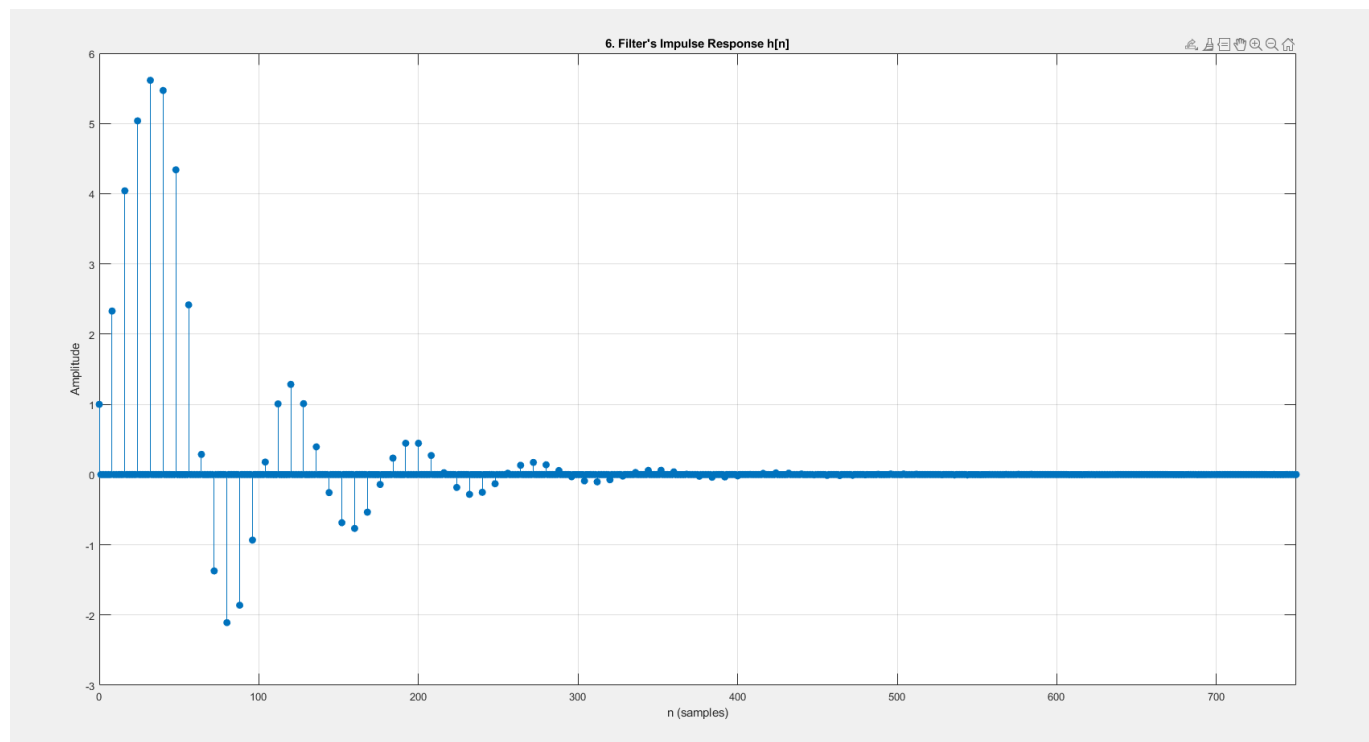
Phase response of the filter in the frequency range $(-\pi < \omega \leq \pi)$ (Using unwrap function)



Group delay of the filter in the frequency range $(-\pi < \omega \leq \pi)$



The filter's impulse response



Very Important note: The code takes some time before all the plots appear due to sweeping loops in the Code.

Appendix (MATLAB Code)

```

1  %% First order discrete-time lowpass filter
2  clear; clc; close all;
3  b = [1];
4  a1 = [1 -0.6];
5  a2 = [1 0.6];
6
7  w = -pi:0.01:pi;
8  z = exp(1j*w);
9
10 %% Pole-zero plot and the unit circle in the Z-domain
11
12 figure('Name', 'First Order LPF - Pole-Zero Plot (a=0.6)');
13 zplane(b,a1);
14 figure('Name', 'First Order LPF - Pole-Zero Plot(a=-0.6)');
15 zplane(b,a2);
16
17 %% Magnitude response of the filter in dB in the frequency range (-pi < w < pi)
18
19 figure('Name', 'First Order LPF - Magnitude Response');
20
21 H1 = 1./(1 - 0.6*(z.^(-1)));
22 H2 = 1./(1 + 0.6*(z.^(-1)));
23
24 plot(w, 20*log10(abs(H1)), 'b', 'Linewidth', 1.5);
25 hold on;
26 plot(w, 20*log10(abs(H2)), 'r', 'Linewidth', 1.5);
27 xlabel('w');
28 ylabel('Magnitude Response (dB)');
29 legend('a = 0.6', 'a = -0.6');
30 grid on;
31
32 %% Phase response of the filter in the frequency range (-pi < w ≤ pi)
33
34 figure('Name', 'First Order LPF - Phase Response');
35 plot(w, angle(H1), 'b', 'Linewidth', 1.5);
36 hold on;
37 plot(w, angle(H2), 'r', 'Linewidth', 1.5);
38 xlabel('w');
39 ylabel('Phase (radian)');
40 legend('a = 0.6', 'a = -0.6');
41 grid on;
42
43 %% Group delay of the filter in the frequency range (-pi < w ≤ pi)
44
45 figure('Name', 'First Order LPF - Group Delay');
46 ph1 = unwrap(angle(H1));
47 ph2 = unwrap(angle(H2));
48
49 gd1 = -diff(ph1)./diff(w);
50 gd2 = -diff(ph2)./diff(w);
51 w_mid = (w(1:end-1) + w(2:end))/2;
52
53 plot(w_mid, gd1, 'b', 'Linewidth', 1.5);
54 hold on;
55 plot(w_mid, gd2, 'r', 'Linewidth', 1.5);
56 xlabel('w');
57 ylabel('Group Delay');
58 legend('a = 0.6', 'a = -0.6');
59 grid on;
60

```




```

61 %% The filter's impulse response
62 [h1, n1] = impz(b, a1);
63 [h2, n2] = impz(b, a2);
64
65 figure('Name', 'First Order LPF - Impulse Response (a=0.6)');
66 stem(n1, h1, 'b');
67 xlabel('n');
68 ylabel('h_1[n]');
69 title('Impulse Response (a = 0.6)');
70 grid on;
71
72 figure('Name', 'First Order LPF - Impulse Response(a=-0.6)');
73 stem(n2, h2, 'r');
74 xlabel('n');
75 ylabel('h_2[n]');
76 title('Impulse Response (a = -0.6)');
77 grid on;
78
79
80
81
82
83
84 %% Third Order Discrete-Time Lowpass Filter
85 clc; clear;
86
87 %% Initial Filter Specifications
88 wc = 0.25*pi; % Cutoff frequency
89 wpo = 0.2*pi; % original passband edge
90 wso = 0.3*pi; % original stopband edge
91
92 w = -pi:0.01:pi;
93
94 % Pole and Zero Placement
95 p1 = 0.6; % Real pole (from first-order LPF)
96
97 %% Sweep on r and wp
98
99 r_values = 0.1 : 0.01 : 0.99; % Range of r
100
101 wp1_values = 0.18*pi : 0.01*pi : 0.22*pi; % Small sweep around 0.2*pi
102
103 optimum_r = 0;
104 optimum_wp1 = 0;
105
106 Optimum_ripple = inf;
107
108 for wp1 = wp1_values
109     for r_trial = r_values
110
111         % Complex poles
112         p2 = r_trial * exp(1j*wp1);
113         p3 = r_trial * exp(-1j*wp1);
114
115         % Complex zeros
116         z2 = exp(1j*wso);
117         z3 = exp(-1j*wso);
118
119         poles_vec = [p1; p2; p3];
120         zeros_vec = [z2; z3];
121
122

```

```

122
123     % Transfer function coefficients
124     b = real(poly(zeros_vec));
125     a = real(poly(poles_vec));
126
127     % Frequency response in passband
128     w_pass = linspace(0, wpo, 1000);
129
130     H_trial = freqz(b, a, w_pass);
131
132     % Magnitude in dB
133     Hdb = 20*log10(abs(H_trial));
134
135     % Passband ripple
136     ripple = max(Hdb) - min(Hdb);
137
138     % Save best parameters
139     if ripple < Optimum_ripple
140
141         Optimum_ripple = ripple;
142         optimum_r = r_trial;
143         optimum_wp1 = wp1;
144
145     end
146 end
147 end
148 %% optimum Parameters
149
150 fprintf('==== Optimum Parameters ==== \n');
151 fprintf('optimum r = %.3f \n', optimum_r);
152 fprintf('optimum wp1 = %.3f*pi \n', optimum_wp1/pi);
153 fprintf('Passband Ripple = %.3f dB \n', Optimum_ripple);
154
155
156 %% The coefficients after finding the suitable r and wp
157
158 p1 = 0.6; % Real pole (from first-order LPF)
159 p2 = optimum_r * exp(1j*optimum_wp1); % Complex pole 1 with optimum r from the sweep and optimum wp
160 p3 = optimum_r * exp(-1j*optimum_wp1); % Complex pole 2 (conjugate) with optimum r from the sweep and optimum wp
161 z2 = exp(1j*wso); % Complex zero 1 on unit circle
162 z3 = exp(-1j*wso); % Complex zero 2 on unit circle
163
164 %% Create transfer function polynomials
165
166 poles = [p1, p2, p3];
167 zeros = [z2, z3];
168
169 B = poly(zeros); % Numerator coefficients
170 A = poly(poles); % Denominator coefficients
171
172 % Frequency Response
173
174 [H, w] = freqz(B, A, 1024, 'whole');
175
176 H_shifted = fftshift(H);
177
178 gd = grpdelay(B, A, w);
179
180

```



```

181 %% 1. Pole-Zero Plot and Unit Circle
182 figure('Name', '3rd Order LPF - Pole-Zero Plot');
183 zplane(B, A);
184 title('Pole-Zero Plot (x: poles, o: zeros)');
185
186 %% 2. Magnitude Response in dB (- $\pi$  to  $\pi$ )
187 figure('Name', '3rd Order LPF - Magnitude Response');
188 gain = max(abs(H_shifted));
189 H_shifted = H_shifted / gain;
190 plot((w-pi)/pi, 20*log10(abs(H_shifted)));
191 xlabel('Normalized Frequency ( $\times\pi$  rad/sample)');
192 ylabel('Magnitude (dB)');
193 title('Fullband Magnitude Response');
194 grid on;
195 xlim([-1, 1]);
196
197 %% 3. Magnitude Response in dB (- $\omega_p$  to  $\omega_p$ ) for Passband Ripple
198 figure('Name', '3rd Order LPF - Passband Detail');
199 plot((w-pi)/pi, 20*log10(abs(H_shifted)));
200 xlabel('Normalized Frequency ( $\times\pi$  rad/sample)');
201 ylabel('Magnitude (dB)');
202 title('Passband Magnitude Response (Zoomed)');
203 grid on;
204 xlim([-wpo/pi, wpo/pi]); % Zoom into the passband
205
206 %% 4. Phase Response in dB (- $\pi$  to  $\pi$ )
207 figure('Name', '3rd Order LPF - Phase Response');
208 plot((w-pi)/pi, unwrap(angle(H_shifted)));
209 xlabel('Normalized Frequency ( $\times\pi$  rad/sample)');
210 ylabel('Phase (radians)');
211 title('Phase Response');
212 grid on;
213 xlim([-1, 1]);
214
215 %% 5. Group Delay Response
216 figure('Name', '3rd Order LPF - Group Delay');
217 plot((w-pi)/pi, fftshift(gd));
218 xlabel('Normalized Frequency ( $\times\pi$  rad/sample)');
219 ylabel('Group Delay (samples)');
220 title('Group Delay Response');
221 grid on;
222 xlim([-1, 1]);
223
224 %% 6. Impulse Response
225 figure('Name', '3rd Order LPF - Impulse Response');
226 impz(B, A);
227 title('Impulse Response');
228 grid on;
229
230
231
232
233
234
235
236 %% Fifth Order Discrete-Time Lowpass Filter- First case
237 clear; clc;
238
239 wc = 0.25 * pi; % Cutoff frequency
240 delta_w = 0.1 * pi; % Transition bandwidth
241 wpo = wc - delta_w/2; % 0.2*pi
242 wso = wc + delta_w/2; % 0.3*pi
243 optimum_wp1 = 0.21*pi; % optimum wp for Ap less than 1 dB for third order
244
245 r1 = 0.88; % Optimum magnitude of poles from third order
246 r2 = 0.6; % real pole from first order
247 r3 = 0.88;
248

```



```

249 %% Zeros (On the unit circle)
250 z1 = exp(1j * wso);
251 z2 = exp(-1j * wso);
252 w_z_order5 = (wso + pi) / 2;
253 z3 = exp(1j * w_z_order5);
254 z4 = exp(-1j * w_z_order5);
255 zeros_vec = [z1; z2; z3; z4];
256
257 %% Poles
258 p1 = r2;
259 p2 = r1 * exp(1j * optimum_wp1);
260 p3 = r1 * exp(-1j * optimum_wp1);
261 w_p_order5 = (0 + optimum_wp1) / 2;
262 p4 = r3 * exp(1j * w_p_order5);
263 p5 = r3 * exp(-1j * w_p_order5);
264 poles_vec = [p1; p2; p3; p4; p5];
265
266 %% Filter order5 Coefficients
267 b = poly(zeros_vec); % Numerator coefficients
268 a = poly(poles_vec); % Denominator coefficients
269
270 %% Full Frequency Response
271 w = -pi:0.005:pi;
272 H_order5_LPF = freqz(b, a, w);
273
274 %% --- Plot 1: Pole-Zero Plot ---
275 figure('Name', '5rd Order LPF - Pole-Zero Plot (Before adjustment)');
276 zplane(b, a);
277 grid on;
278 title('1. Pole-Zero Plot (Order 5)');
279
280 %% Plot 2: Full Magnitude Response
281 figure('Name', '5rd Order LPF - Magnitude Response (Before adjustment)');
282 plot(w, 20*log10(abs(H_order5_LPF)));
283 xlabel('\omega (rad/sample)');
284 ylabel('Magnitude Response (dB)');
285 title('2. Full Magnitude Response (-\pi < \omega \leq \pi)');
286 grid on;
287
288 %% Plot 3: Passband Magnitude Response (For Ripple Calculation)
289 figure('Name', '5rd Order LPF - Passband Detail');
290 w_passband = -wpo:0.001:wpo;
291 H_passband = freqz(b, a, w_passband);
292 plot(w_passband, 20*log10(abs(H_passband)));
293 xlabel('\omega (rad/sample)');
294 ylabel('Magnitude Response (dB)');
295 title('3. Passband Magnitude Response (-\omega_p \leq \omega \leq \omega_p)');
296 grid on;
297
298 %% Plot 4: Phase Response
299 figure('Name', '5rd Order LPF - Phase Response (Before adjustment)');
300 plot(w, unwrap(angle(H_order5_LPF)));
301 xlabel('\omega (rad/sample)');
302 ylabel('Phase (radians)');
303 title('4. Phase Response (-\pi < \omega \leq \pi)');
304 grid on;
305

```



```

306 %% Plot 5: Group Delay Response
307 figure('Name', '5rd Order LPF - Group Delay Response (Before adjustment)');
308 [gd, w_gd] = grpdelay(b, a, w);
309 plot(w_gd, gd);
310 xlabel('\omega (rad/sample)');
311 ylabel('Group Delay (samples)');
312 title('5. Group Delay Response ( $-\pi < \omega \leq \pi$ )');
313 grid on;
314
315 %% --- Plot 6: Impulse Response ---
316 figure('Name', '5rd Order LPF - Impulse Response (Before adjustment)');
317 impz(b, a);
318 title('6. Filter''s Impulse Response h[n]');
319 grid on;
320
321
322 %% Sweep on r1, r2 and r3
323
324 r1_values = 0.1 : 0.01 : 0.99;
325 r2_values = 0.1 : 0.01 : 0.99;
326 r3_values = 0.1 : 0.01 : 0.99;
327
328 Optimum_r1 = 0;
329 Optimum_r2 = 0;
330 Optimum_r3 = 0;
331 Optimum_ripple = inf;
332
333 for r1_trial = r1_values
334     for r2_trial = r2_values
335         for r3_trial = r3_values
336
337             % Complex Zeros
338             z1 = exp(1j*wso);
339             z2 = exp(-1j*wso);
340             w_z_order5 = (wso + pi)/2;
341             z3 = exp(1j*w_z_order5);
342             z4 = exp(-1j*w_z_order5);
343
344             zeros_vec = [z1; z2; z3; z4];
345
346             % Poles
347             p1 = r2_trial;
348             p2 = r1_trial * exp(1j*optimum_wp1);
349             p3 = r1_trial * exp(-1j*optimum_wp1);
350             w_p_order5 = (0 + optimum_wp1)/2;
351             p4 = r3_trial * exp(1j*w_p_order5);
352             p5 = r3_trial * exp(-1j*w_p_order5);
353
354             poles_vec = [p1; p2; p3; p4; p5];
355
356             % Transfer Function Coefficients
357             b = real(poly(zeros_vec));
358             a = real(poly(poles_vec));
359
360             % Frequency Response in Passband
361             w_passband = linspace(0, wpo, 2000);
362
363             H_trial = freqz(b, a, w_passband);
364
365             % Normalize Gain
366             H_trial = H_trial / abs(H_trial(1));
367
368             % Magnitude in dB
369             Hdb = 20*log10(abs(H_trial));
370
371             % Passband Ripple
372             ripple = max(Hdb) - min(Hdb);
373
374             % Save Optimum Parameters
375
376
377

```

```

377
378         if ripple < Optimum_ripple
379
380             Optimum_ripple = ripple;
381
382             Optimum_r1 = r1_trial;
383             Optimum_r2 = r2_trial;
384             Optimum_r3= r3_trial;
385         end
386
387     end
388 end
389 end
390
391 % Optimum Parameters
392
393 fprintf('==== Optimum Parameters =====\n');
394
395 fprintf('Optimum r1 = %.3f\n', Optimum_r1);
396 fprintf('Optimum r2 = %.3f\n', Optimum_r2);
397 fprintf('Optimum r3 = %.3f\n', Optimum_r3);
398 fprintf('Passband Ripple = %.3f dB\n', Optimum_ripple);
399
400
401
402
403
404 %% Fifth Order Discrete-Time Lowpass Filter- Second case with optimum r1, r2 and r3 with passband ripples = 0.444 dB
405
406 r1 = 0.9;
407 r2 = 0.19;
408 r3 = 0.52;
409
410 %% Zeros (On the unit circle)
411 z1 = exp(1j * wso);
412 z2 = exp(-1j * wso);
413 w_z_order5 = (wso + pi) / 2;
414 z3 = exp(1j * w_z_order5);
415 z4 = exp(-1j * w_z_order5);
416 zeros_vec = [z1; z2; z3; z4];
417
418 %% Poles
419 p1 = r2;
420 p2 = r1 * exp(1j * optimum_wp1);
421 p3 = r1 * exp(-1j * optimum_wp1);
422 w_p_order5 = (0 + optimum_wp1) / 2;
423 p4 = r3 * exp(1j * w_p_order5);
424 p5 = r3 * exp(-1j * w_p_order5);
425 poles_vec = [p1; p2; p3; p4; p5];
426
427 %% Filter order5 Coefficients
428 b = poly(zeros_vec); % Numerator coefficients
429 a = poly(poles_vec); % Denominator coefficients
430
431 %% Full Frequency Response
432 w = -pi:0.005:pi;
433 H_order5_LPF = freqz(b, a, w);
434
435 %% --- Plot 1: Pole-Zero Plot ---
436 figure('Name', '5rd Order LPF - Pole-Zero Plot (After adjustment)');
437 zplane(b, a);
438 grid on;
439 title('1. Pole-Zero Plot (Order 5)');

```



```

441 %% Plot 2: Full Magnitude Response
442 figure('Name', '5rd Order LPF - Magnitude Response (After adjustment)');
443 plot(w, 20*log10(abs(H_order5_LPF)));
444 xlabel('\omega (rad/sample)');
445 ylabel('Magnitude Response (dB)');
446 title('2. Full Magnitude Response ( $-\pi < \omega \leq \pi$ )');
447 grid on;
448
449 %% Plot 3: Passband Magnitude Response (For Ripple Calculation)
450 figure('Name', '5rd Order LPF - Passband Detail');
451 w_passband = -wpo : 0.001 : wpo;
452 H_passband = freqz(b, a, w_passband);
453 plot(w_passband, 20*log10(abs(H_passband)));
454 xlabel('\omega (rad/sample)');
455 ylabel('Magnitude Response (dB)');
456 title('3. Passband Magnitude Response ( $-\omega_p \leq \omega \leq \omega_p$ )');
457 grid on;
458
459 %% Plot 4: Phase Response
460 figure('Name', '5rd Order LPF - Phase Response (After adjustment)');
461 plot(w, unwrap(angle(H_order5_LPF)));
462 xlabel('\omega (rad/sample)');
463 ylabel('Phase (radians)');
464 title('4. Phase Response ( $-\pi < \omega \leq \pi$ )');
465 grid on;
466
467 %% Plot 5: Group Delay Response
468 figure('Name', '5rd Order LPF - Group Delay Response (After adjustment)');
469 [gd, w_gd] = grpdelay(b, a, w);
470 plot(w_gd, gd);
471 xlabel('\omega (rad/sample)');
472 ylabel('Group Delay (samples)');
473 title('5. Group Delay Response ( $-\pi < \omega \leq \pi$ )');
474 grid on;
475
476 %% -- Plot 6: Impulse Response ---
477 figure('Name', '5rd Order LPF - Impulse Response (After adjustment)');
478 impz(b, a);
479 title('6. Filter''s Impulse Response h[n]');
480 grid on;
481
482
483
484
485
486
487
488 %% Frequency Transformation using Pole-Zero Pattern Rotation (HPF)
489
490 zeros_HPF = -zeros_vec; % Rotate all zeros by 180° on the z-plane
491 poles_HPF = -poles_vec; % Rotate all poles by 180° on the z-plane
492
493 b_HPF = poly(zeros_HPF);
494 a_HPF = poly(poles_HPF);
495
496 %% Full Frequency Response
497 w = -pi:0.005:pi;
498 H_order5_HPF = freqz(b_HPF, a_HPF, w);
499
500 %% --- Plot 1: Pole-Zero Plot ---
501 figure('Name', '5rd Order HPF - Pole-Zero Plot');
502 zplane(b_HPF, a_HPF);
503 grid on;
504 title('7. Pole-Zero Plot of the HPF');
505
506 %% Plot 2: Full Magnitude Response
507 figure('Name', '5rd Order HPF - Magnitude Response ');
508 plot(w, 20*log10(abs(H_order5_HPF)));
509 xlabel('\omega (rad/sample)');

```



```

510 ylabel('Magnitude Response (dB)');
511 title('2. Full Magnitude Response ( $-\pi < \omega \leq \pi$ )');
512 grid on;
513
514 %% Plot 3: Passband Magnitude Response (For Ripple Calculation)
515 figure('Name', '5rd Order HPF - Passband Magnitude Response ');
516 w_passband = pi-wpo:0.001:pi ;
517 H_passband = freqz(b_HPF, a_HPF, w_passband);
518 plot(w_passband, 20*log10(abs(H_passband)));
519 xlabel('\omega (rad/sample)');
520 ylabel('Magnitude Response (dB)');
521 title('3. Passband Magnitude Response');
522 grid on;
523
524 %% Plot 4: Phase Response
525 figure('Name', '5rd Order HPF - Phase Response');
526 plot(w, unwrap(angle(H_order5_HPF)));
527 xlabel('\omega (rad/sample)');
528 ylabel('Phase (radians)');
529 title('4. Phase Response ( $-\pi < \omega \leq \pi$ )');
530 grid on;
531
532 %% Plot 5: Group Delay Response
533 figure('Name', '5rd Order HPF - Group Delay Response');
534 [gd, w_gd] = grpdelay(b_HPF, a_HPF, w);
535 plot(w_gd, gd);
536 xlabel('\omega (rad/sample)');
537 ylabel('Group Delay (samples)');
538 title('5. Group Delay Response ( $-\pi < \omega \leq \pi$ )');
539 grid on;
540
541 %% -- Plot 6: Impulse Response ---
542 figure('Name', '5rd Order HPF - Impulse Response');
543 impz(b_HPF, a_HPF);
544 title('6. Filter''s Impulse Response h[n]');
545 grid on;
546
547
548
549 %% Frequency Transformation using Pole-Zero Pattern Rotation (BPF)
550
551 % Apply Frequency Transformation (BPF at  $\pi/2$ )
552 % To ensure real coefficients, we must include BOTH  $+\pi/2$  and  $-\pi/2$  shifts.
553 % This doubles the order of the filter
554
555 shift_pos = exp(1j * pi/2);
556 shift_neg = exp(-1j * pi/2);
557
558 zeros_vec_bpf = [zeros_vec * shift_pos; zeros_vec * shift_neg];
559 poles_vec_bpf = [poles_vec * shift_pos; poles_vec * shift_neg];
560
561 % Filter BPF Coefficients
562 b_bpf = real(poly(zeros_vec_bpf));
563 a_bpf = real(poly(poles_vec_bpf));
564
565 %% --- 4. Plotting ---
566 w = -pi:0.005:pi;
567 H_bpf = freqz(b_bpf, a_bpf, w);
568
569 %% Plot 1: Pole-Zero Plot
570 figure('Name', '5rd Order BPF - Pole-Zero Plot');
571 zplane(b_bpf, a_bpf);
572 grid on;
573 title('1. Pole-Zero Plot (10th Order BPF)');
574
575 %% Plot 2: Full Magnitude Response
576 figure('Name', '5rd Order BPF - Magnitude Response');
577 plot(w, 20*log10(abs(H_bpf)));
578 xlabel('\omega (rad/sample)');

```



```

579 ylabel('Magnitude Response (dB)');
580 title('2. Full Magnitude Response ( $-\pi < \omega \leq \pi$ )');
581 grid on;
582
583 %% Plot 3: Passband Magnitude Response (For Ripple Calculation)
584 % The passband is now centered around  $\pi/2$  instead of 0.
585 figure('Name', '5rd Order BPF - Passband Magnitude Response');
586 w_center = pi/2;
587 w_pass_bpf = (w_center - wpo):0.001:(w_center + wpo);
588 H_pass_bpf = freqz(b_bpf, a_bpf, w_pass_bpf);
589
590 plot(w_pass_bpf, 20*log10(abs(H_pass_bpf)));
591 xlabel('\omega (rad/sample)');
592 ylabel('Magnitude Response (dB)');
593 title('3. Passband Magnitude Response (BPF Passband)');
594 grid on;
595
596 %% Plot 4: Phase Response
597 figure('Name', '5rd Order BPF - Phase Response');
598 plot(w, unwrap(angle(H_bpf)));
599 xlabel('\omega (rad/sample)');
600 ylabel('Phase (radians)');
601 title('4. Phase Response ( $-\pi < \omega \leq \pi$ )');
602 grid on;
603
604 %% Plot 5: Group Delay Response
605 figure('Name', '5rd Order BPF - Group Delay Response');
606 [gd, w_gd] = grpdelay(b_bpf, a_bpf, w);
607 plot(w_gd, gd);
608 xlabel('\omega (rad/sample)');
609 ylabel('Group Delay (samples)');
610 title('5. Group Delay Response ( $-\pi < \omega \leq \pi$ )');
611 grid on;
612
613 %% Plot 6: Impulse Response
614 figure('Name', '5rd Order BPF - Impulse Response');
615 impz(b_bpf, a_bpf);
616 title('6. Filter''s Impulse Response h[n] (BPF)');
617 grid on;
618
619
620
621
622
623
624
625 %% Comb Filter
626 clear; clc;
627
628 % Optimum magnitude values from fifth order
629 r1 = 0.9 ;
630 r2 = 0.19 ;
631 r3 = 0.52 ;
632
633
634 wc = 0.25 * pi;           % Cutoff frequency
635 delta_w = 0.1 * pi;      % Transition bandwidth
636 wpo = wc - delta_w/2;     % 0.2*pi
637 wso = wc + delta_w/2;     % 0.3*pi
638 optimum_wp1 = 0.21*pi ;  % optimum wp for Ap less than 1 dB for third order
639
640
641 %% Zeros (On the unit circle)
642 z1 = exp(1j * wso);
643 z2 = exp(-1j * wso);
644 w_z_order5 = (wso + pi) / 2;
645 z3 = exp(1j * w_z_order5);

```



```

646 z4 = exp(-1j * w_z_order5);
647 zeros_vec = [z1; z2; z3; z4];
648
649 %% Poles
650 p1 = r2;
651 p2 = r1 * exp(1j * optimum_wp1);
652 p3 = r1 * exp(-1j * optimum_wp1);
653 w_p_order5 = (0 + optimum_wp1) / 2;
654 p4 = r3 * exp(1j * w_p_order5);
655 p5 = r3 * exp(-1j * w_p_order5);
656 poles_vec = [p1; p2; p3; p4; p5];
657
658 %% Filter Coefficients
659 b = poly(zeros_vec); % Numerator coefficients
660 a = poly(poles_vec); % Denominator coefficients
661
662 L = 8; % Comb filter factor
663
664 b_comb = zeros(1, (length(b)-1)*L + 1); % New numerator coefficients
665 a_comb = zeros(1, (length(a)-1)*L + 1); % New denominator coefficients
666
667 b_comb(1:L:end) = b; % Insert zeros between numerator coefficients
668 a_comb(1:L:end) = a; % Insert zeros between denominator coefficients
669
670 %% Full Frequency Response
671 w = -pi:0.005:pi;
672 H_comb = freqz(b_comb, a_comb, w);
673
674 %% --- Plot 1: Pole-Zero Plot ---
675 figure('Name', ' Comb Filter - Pole-Zero Plot');
676 zplane(b_comb, a_comb);
677 grid on;
678 title('1. Pole-Zero Plot (Comb Filter)');
679
680 %% Plot 2: Full Magnitude Response
681 figure('Name', ' Comb Filter - Magnitude Response');
682 plot(w, 20*log10(abs(H_comb)));
683 xlabel('\omega (rad/sample)');
684 ylabel('Magnitude Response (dB)');
685 title('2. Full Magnitude Response (-\pi < \omega \leq \pi)');
686 grid on;
687
688 %% Plot 3: Passband Magnitude Response
689 figure('Name', ' Comb Filter - Passband Magnitude Response');
690 w_pass_comb = -wpo:0.001:wpo;
691 H_passband = freqz(b, a, w_pass_comb);
692 plot(w_pass_comb, 20*log10(abs(H_passband)));
693 xlabel('\omega (rad/sample)');
694 ylabel('Magnitude Response (dB)');
695 title('3. Passband Magnitude Response (-\omega_p \leq \omega \leq \omega_p)');
696 grid on;
697
698 %% Plot 4: Phase Response
699 figure('Name', ' Comb Filter - Phase Response');
700 plot(w, unwrap(angle(H_comb)));
701 xlabel('\omega (rad/sample)');
702 ylabel('Phase (radians)');
703 title('4. Phase Response (-\pi < \omega \leq \pi)');
704 grid on;
705
706 %% Plot 5: Group Delay Response
707 figure('Name', ' Comb Filter - Group Delay Response');
708 [gd, w_gd] = grpdelay(b_comb, a_comb, w);
709 plot(w_gd, gd);
710 xlabel('\omega (rad/sample)');
711 ylabel('Group Delay (samples)');
712 title('5. Group Delay Response (-\pi < \omega \leq \pi)');
713 grid on;
714
715 %% -- Plot 6: Impulse Response ---
716 figure('Name', ' Comb Filter - Impulse Response');
717 impz(b_comb, a_comb);
718 title('6. Filter''s Impulse Response h[n]');
719 grid on;
720

```